

Personality is a person-specific complex, nonlinear dynamic system

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Authors' note. Correspondence concerning this article should be addressed to Emorie D Beck, One Shields Ave., Davis, CA 95616. Email: edbeck@ucdavis.edu. Emorie Beck and Adam Nissen were supported by a National Institute of Aging Grant RF1AG088206 (PI: Beck), and Emorie Beck and Elizabeth Long were supported by a National Science Foundation Grant BCS-2336406 (co-PI: Beck).

Acknowledgements. This chapter is dedicated to Randy Larsen (1963-2022), whose pioneering work on psychological dynamics was foundational to the field and whose guidance helped shape the ideas in this chapter. We further acknowledge Jack C. Wright, without whose scholarship, mentorship, and longstanding commitment to idiographic personality science this chapter would not have been possible. We are grateful to members of the Beck Lab community (Winkie Ma, Evan Warfel, Rocci Vizzusi, Rongxin Cheng, Anabel Büchner, and Rohit Batra) for years of discussion, feedback, and intellectual engagement that sharpened many of the ideas developed here. We are additionally grateful to Bill Revelle and Dan McAdams for thoughtful feedback on an earlier version of this manuscript.

Abstract

Psychology aims to describe, predict, and explain what people think, feel, and do at both individual and group levels. In personality psychology, this dual aim has historically appeared in broad population-level trait models alongside person-specific studies of the whole person. Yet as trait approaches have become dominant, traits have become nearly synonymous with personality itself. Drawing on historical, epistemological, formal, and empirical analyses, we examine the ontological and practical limitations of this population-based paradigm and consider the promise of a personality science that treats the individual as its primary unit of analysis. We argue that personality is best understood as a person-specific complex, nonlinear dynamic system: A multi-level organization of thoughts, feelings, behaviors, and biological, situational, and cultural forces evolving through reciprocal interactions. We reconceptualize trait structures as emergent summaries of person-specific dynamics rather than primary causal units and outline a bottom-up roadmap for integrative personality science.

Keywords: person-specific processes; dynamic systems; idiographic; personality traits; emergence; nonlinear dynamics; whole person; population-level regularities; personality theory; bottom-up integration

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1 Introduction

Psychology aims to describe, predict, and explain what people think, feel, and do, both at the level of individuals and groups. This dual aim requires identifying general principles of psychological functioning while capturing the nuances of individual persons (e.g., Kuper et al., 2024; Modersitzki et al., 2025; Phan et al., 2025). However, over the past century, psychology has struggled to fully achieve these goals. The discipline has prioritized population-level findings over insights into individuals, leaving a critical gap in our understanding of the person (e.g., Beck & Jackson, 2020b). Population-level aggregates obscure individuality and rarely reflect general principles that apply across individuals (Moeller, 2021; Phan et al., 2025).

In personality psychology, the dual aim of population- versus person-level findings has taken the form of broad, population-level traits versus narrow, person-specific studies of the whole person. Over the last century, the trait-based paradigm has become the science of personality, at least as it is perceived both within and across fields, with traits becoming synonymous with personality (Rauthmann, 2024). Yet this chapter is not a victory lap for traits. It asks a more uncomfortable question: *Did we follow the wrong path?* To address this, we draw on historical, epistemological, formal, and empirical analyses to examine the ontological and practical limitations of the population-based paradigm and explore the promises of a future psychology that takes the individual more seriously as its core unit of analysis.

This question is not merely philosophical. It emerges from a growing body of empirical work made possible by advances in intensive longitudinal assessment, person-specific modeling, and dynamical systems methods, which together now permit direct study of personality as it unfolds within individuals across time. Across studies of person-specific structure, dynamics,

change, predictive heterogeneity, and bottom-up integration (e.g., Beck et al., 2025, 2026; Beck & Jackson, 2020a; Hawks et al., 2024; C. J. Lee & Beck, 2025a; Long, Elsaadawy, et al., 2026; Nissen & Beck, 2025b, 2025a), a consistent conclusion has begun to emerge: Population-level trait models are poorly matched to the multivariate, nonlinear, and person-specific phenomena they are routinely asked to explain (e.g., Beck et al., 2023; Beck & Jackson, 2020b; Kuper et al., 2025; Modersitzki et al., 2025; Phan et al., 2025). Much of this evidence has come from our own coordinated program of research on person-specific personality dynamics. Accordingly, the present chapter serves not only as a conceptual intervention, but as an integrative synthesis of a broader empirical literature and research program aimed at understanding how personality is organized within persons and whether, and in what form, such organization may generalize across them.

Ultimately, we argue that personality is best understood as a person-specific complex, nonlinear dynamic system that operates across levels of aggregation and timescales – that is, as an organized system within persons whose patterns of stability and change emerge through ongoing interactions among psychological, biological, situational, and cultural processes. By stating that personality “is” a person-specific nonlinear dynamic system, we are not asserting this by fiat. Rather, we argue that given a set of widely endorsed criteria for what personality must explain, a dynamic systems ontology provides the most coherent and complete account currently available. This chapter does not introduce another theory into an already crowded landscape. Instead, we advance a framework for understanding personality that organizes how phenomena are defined and studied across levels of analysis, rather than proposing a new set of constructs. Our goal is not to replace one theory with another, but to articulate a more adequate foundation for building and integrating theories. After reviewing evidence and laying out arguments for how

and why personality is best understood as a dynamic system, we conclude with a roadmap for a new science of personality, and psychology more broadly, that starts with the single case and considers whether and when general principles may emerge.

2 Historical: Personality as the Study of the Whole Person

As noted above, psychology aims to describe, predict, and explain the things that people think, feel, and do, for one person, some people, and all people. Yet the bulk of scholarship in personality over the last century has focused on so-called trait approaches to personality. Traits are “characteristics of the person that account for consistent patterns of feelings, thinking, and behaving” (Pervin et al., 2005, p. 6), typically operationalized as relative differences on broad, hierarchical latent dimensions that differentiate core characteristics of populations. This dominance can obscure that personality psychology has long had two partially competing tasks: to describe how people differ from one another on shared attributes and to describe and explain the organization of individuals holistically across time and contexts (Cervone, 1991; Winter & Barenbaum, 1999). Understanding how the field moved from this dual agenda to the contemporary trait paradigm is essential not only for explaining personality psychology’s current structure, but also for clarifying the directions in which it can evolve. A full history could span volumes, so we focus on several key intellectual developments that progressively separated person-specific and population-level approaches and helped solidify traits as personality psychology’s primary unit of analysis.

2.1 Two Sciences of Personality

The first of these tasks focused on understanding what underlies the *stability* of behavior across people, time, and situations. The second meant to understand the *dynamics* and variability of behavior within a single person across time and situations (i.e., the whole person). The

idiographic-nomothetic distinction was reflected in methodological differences within the field, still seen today in the tension between qualitative and quantitative approaches. While nomothetic approaches have relied on quantitative models, idiographic approaches historically primarily existed in psychobiography and clinical work (McAdams, 1997; Winter & Barenbaum, 1999).

Idiographic and nomothetic approaches have generally been regarded as conceptually distinct (Block, 1995; Cervone, 2005) but whether they are truly distinct – or instead reflect different levels of analysis connected by unknown mechanisms – has not been adequately addressed. While theoretical arguments on the distinction between the dynamics of the single case and differences across people have abounded (e.g., Allport, 1937; Beck & Jackson, 2020; Borsboom et al., 2003; Cattell, 1946; Cervone, 2005; Molenaar, 2004; Murray, 1938; Shoda et al., 1993; Stern, 1917; Wright & Woods, 2020), empirical tests merging the idiographic and the nomothetic have largely remained elusive. Recent methodological advances, however, have allowed for an explosion of idiographic scholarship. Merged with the growing literature on personality dynamics, we argue that the study of *idiographic personality dynamics* will finally shed light on centuries old questions about distinctions between idiographic and nomothetic approaches (e.g., Beck & Jackson, 2021a, 2021c).

The distinction between nomothetic and idiographic approaches is complex¹. For (approximate) consistency with their use in debates at the time, the current historical section uses *idiographic* to refer to research where the individual is the focal entity, wherein the inferential goal is to generalize to that person across time, and *nomothetic* to refer to research on population- and group-level differences, wherein the inferential goal is to generalize across

¹ Notably, the usage of these terms as introduced above is a divergence from their original introduction by Windelband (1894). As detailed in Lamiell (1998), Windelband never meant the distinction to apply to the difference between person(s) and group(s). A more thorough treatment of the historical and modern uses of the terms nomothetic and idiographic can also be found in Phan et al. (2025).

people. However, each has been used to refer to a wide variety of inferential goals and research entities (Phan et al., 2025), rendering them less than ideal as a means of classifying research programs. Thus, throughout the rest of the chapter, we instead use person-specific and population-level as more accurate descriptors.

The rest of this section proceeds by charting the intellectual history of personality's two tasks beginning with William Stern (1900) and continuing to the present day.

2.2 Personality and the Whole Person

Early work in personality at the turn of the 20th century was rooted in the task of understanding the whole person. At the same time, personality psychology sought to establish itself as an empirical science by defining its units of analysis (i.e., how personality is studied). Influenced by developments in natural taxonomy (e.g., the periodic table, biological classification; Lamiell, 2003), clinical assessment (Kraepelin, 1904), intelligence testing (Binet & Simon, 1905; Stern, 1912), and the rise of factor analysis and psychometrics (Cattell, 1946b; Spearman, 1904), the field confronted a central tension: Conceptualizations emphasized the idiographic, integrated person, whereas empirical approaches prioritized nomothetic differentiation.

Although Gordon Allport is commonly considered the “father” of personality in the United States, William Stern's earlier work on personality (1906, 1911) and individual differences (1900, 1911) predates Allport by several decades. Stern's (1911, 1918) critical personalism rests on a fundamental ontological distinction between persons and things, where persons are unities in multiplicity (*unitas multiplex*) – irreducible wholes whose parts are organized into meaningful, value-laden, and purposive life structures rather than mere aggregates. This philosophical framework provides a holistic account of persons as organized,

goal-directed systems in dynamic relation with their environments, where behavior reflects the interplay of internal tendencies and external conditions rather than reducible components or isolated causes, rejecting reductionist and mechanistic accounts.

2.3 The Unit of Personality

Gordon Allport helped establish personality psychology as an empirical discipline, particularly by advancing traits as a unit of personality. Allport and Odbert (1936) introduced the lexical hypothesis, arguing that individual differences can be identified through everyday language. Their compilation of nearly 18,000 trait descriptors helped solidify traits as central to personality research (G. W. Allport & Odbert, 1936; John, 2021).

At the same time, Allport's own conceptualization of personality was not reducible to traits. He defined personality as "the dynamic organization within the individual of those psychophysical systems that determine his unique adjustments to the environment" (Allport, 1937, p. 48). To Allport, so-called "common traits" (i.e., between-person) were merely useful abstractions. Instead, he believed that the organizing units of personality were better captured by person-specific "individual traits" whose content and operation emerged from individuals' unique meaning-making processes (Allport, 1937, pp. 297). These individual traits have a distinctive organization within each person, such that most people could be understood as having a set of core "cardinal" or "central traits" that acted as unique organizing foci for the system, as well as more circumscribed "secondary traits" that supported their unfolding within specific situations. Thus, despite his role in the development of the common trait framework that dominates modern personality psychology, Allport's vision of how traits actually operate within an individual diverges substantially: (1) the content and operation of traits within a given individual diverges from what is described in common trait frameworks due to the specifics of

one's life circumstances and meaning-making; (2) a person's traits must be understood in relation to each other and to one's circumstances rather than as isolated dimensions or modules; and (3) traits on their own do not capture the whole of personality. This aspect of his work shone most in his idiographic application of the lexical approach, as in his analysis of *Letters from Jenny*, using ratings and early text analysis to identify person-specific structures (G. W. Allport, 1965).

Despite this, Allport is largely remembered for his role in establishing common traits, a legacy that obscures his broader, more integrative vision for personality science (Beck & Jackson, 2020b; Long, Elsaadawy, et al., 2026; Zuroff, 1986).

2.4 Personality Structure and the Rise of the Big Few

Despite Allport's critiques of the limits of nomothetic, common traits, multiple historical developments converged to solidify them as *the* unit of personality. Most critically, the need for large-scale screening, selection, and placement, especially in military and later occupational settings (Gibby & Zickar, 2008), coupled with the introduction of factor analysis and psychometrics, transformed not only personality psychology but psychology more broadly. Factor analysis provided a quantitative framework for data reduction (i.e., identifying latent dimensions underlying high-dimensional data), aligning naturally with traits as either causal forces of narrower indicators or as broad summaries of them (Cattell, 1943).

These techniques also formalized a question that remains central to the field: What is the most parsimonious set of broad traits that best differentiate individuals or populations? Within this context, Raymond B. Cattell (1943, 1946b, 1946a) played a pivotal role in advancing factor analytic approaches and articulating a broader framework for psychological measurement. His data box (see Figure 1) conceptualized data across three dimensions: persons (P), variables (X),

and occasions (T). Critically, the questions psychology can answer depend on how this space is oriented and aggregated.

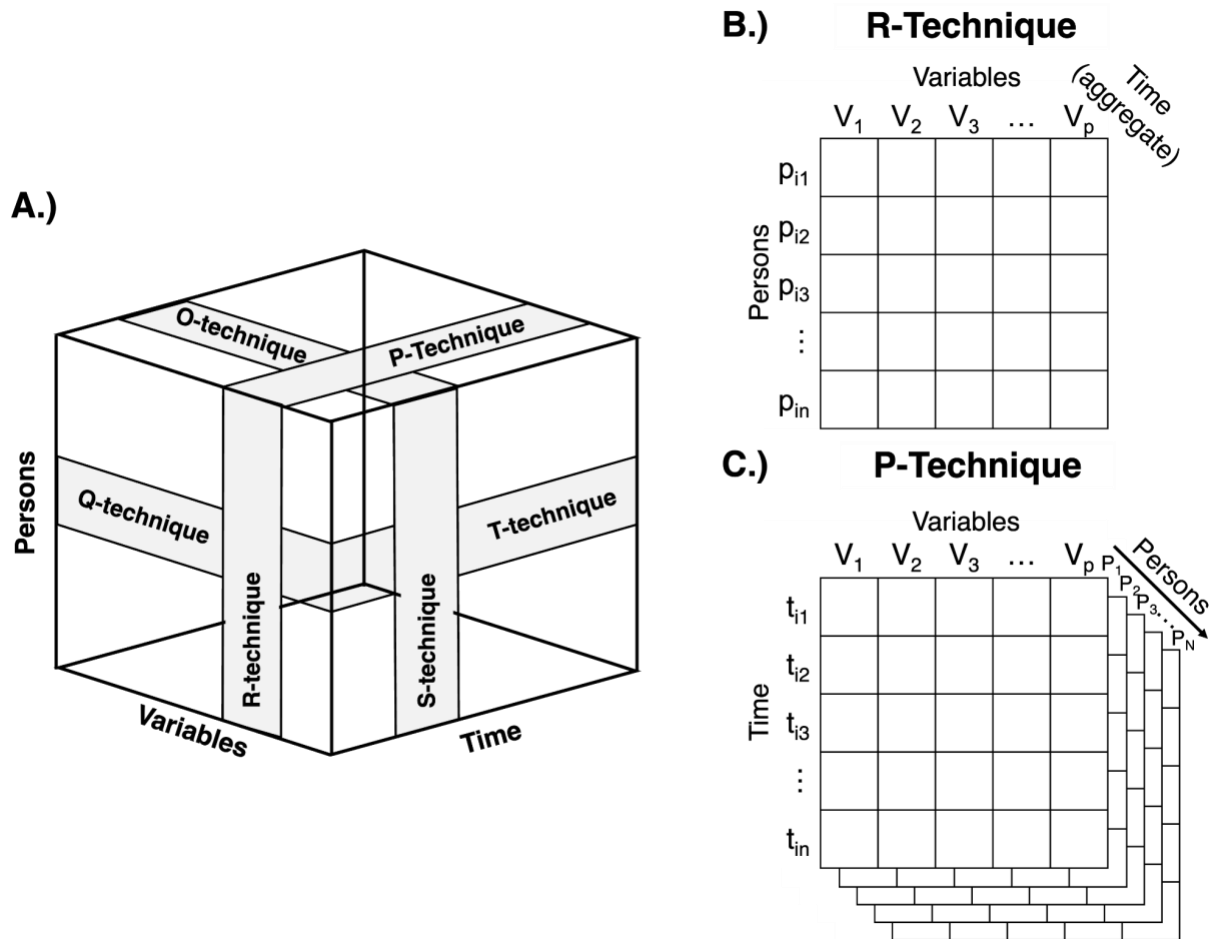


Figure 1. (A) Cattell's (1946b) data box indexing variables (V), persons (P), and time (T). Each orientation corresponds to a distinct mode of aggregation, represented by grey banners perpendicular to the dimension being clustered. These orientations reflect fundamentally different inferential targets depending on how variation is organized across persons and time. (B) The R-technique, which identifies clusters of variables (V) across persons (P), typically after aggregating observations across time, thereby emphasizing population-level structure. (C) The P-technique, which identifies clusters of variables (V) across time (T) within individuals, by slicing along the person dimension (P) to estimate person-specific structures.

For example, R-technique reduces variables (X) to latent dimensions that best differentiate persons (P), typically aggregating across time (T), and thus captures between-person common traits. In contrast, P-technique reduces variables to dimensions that differentiate occasions (T) within a single person (P), identifying unique traits. Other orientations, such as Q-

technique, identify typologies of individuals with similar response patterns, seen today in approaches like latent profile analysis (e.g., Tein et al., 2013).

Importantly, Cattell (1946a) viewed these approaches as complementary, arguing that whether within- and between-person structures are equivalent is an empirical question that he suspected would not resolve in favor of equivalence. Yet, large-scale tests of this question waned for decades due to the absence of intensive longitudinal data necessary for capturing all orientations of the data box at scale.

2.5 The Rise of Systems Thinking

The mid-20th century saw the emergence of dynamical systems approaches across the sciences (e.g., information theory, cybernetics; Shannon, 1948; Wiener & Von Neumann, 1949), alongside growing dissatisfaction with the mechanistic assumptions underlying behaviorism. Within and beyond psychology, scholars increasingly recognized that system-level properties, such as equilibrium, generalize beyond physical and biological systems.

In Ludwig von Bertalanffy's (1950) general systems theory, living organisms were characterized by organized complexity and understood as open systems that maintain structure through continual exchange with their environments. Related ideas quickly entered psychological thinking, including Ross Stagner's (1951) proposal of homeostasis as a unifying principle across social and psychological systems.

Gordon Allport (1960) explicitly applied system concepts to the domain of personality psychology, particularly the distinction between closed and open systems. He argued that psychological theories differ in the extent to which they reflect core properties of open systems, including energy exchange, homeostasis, increasing order over time, and the relationship between person and environment.

In our view, the most consequential of these is the final criterion: Whether personality is something “within the skin” or something that emerges through interaction with the environment. The former aligns with trait-based perspectives, whereas the latter demands an account of how persons and environments dynamically co-constitute one another.

Allport’s core argument was that psychology’s central task is to understand the “person-system,” of which traits are only one component. Most critically, he ultimately argued that seeking generalities necessarily imposes a quasi-closed system outlook, while an open systems perspective requires an idiographic outlook. He concludes, “Let me repeat this question, for it is the one that more than any other has haunted me over the years. *What makes the system cohere in any one person?* [emphasis added]” (Allport, 1960, p. 308). This question haunts these authors as well, and it is one we will return to for the bulk of the second half of the chapter.

2.6 The Person-Situation Debate

The introduction of factor analysis sparked a race to identify the most parsimonious set of dimensions that could differentiate populations (e.g., Cattell and Eber’s [1950] 16 personality factors; Eysenck and Eysenck’s [1965] 3-dimension PEN model; Norman’s [1963] “Adequate Taxonomy”; Tellegen and Waller’s [1995] Big Seven model). At the same time, systems thinking continued to foreground tensions between idiographic and nomothetic approaches, while also pointing to their potential integration. That possibility began to collapse in 1968 with Walter Mischel’s *Personality and Assessment* (Mischel, 1968).

That the book had deleterious impacts on personality psychology is uncontested. Appearing just as the field was consolidating around trait-based approaches, it was widely received as a fundamental challenge to the legitimacy of personality itself and helped precipitate a period of disciplinary uncertainty. Mischel demonstrated that trait variables, while temporally

stable, were cross-situationally inconsistent and only modestly predictive of behavior ($r \approx .30$). He argued that situational variables possessed comparable predictive utility, and therefore that *traits*, particularly when conceptualized as aggregated, decontextualized between-person differences, were insufficient as explanatory mechanisms. His target was not personality per se, but traits as the core explanatory unit.

At the same time, the traits Mischel critiqued did not fully reflect Allport's original vision. Allport's traits were idiographic, embedded within the organization of the individual, and explicitly expected to vary across time and situations (Allport, 1937). In this sense, the debate was not simply between traits and situations, but between two incomplete formulations: One emphasizing consistency without adequately modeling variability, and the other emphasizing variability while rejecting the necessity of traits. Critically, neither framework resolved the problem of linking processes within individuals to differences across individuals.

Rather than integrating these perspectives, the field began to separate them. On one side, trait-based approaches came to dominate, focusing on the structure, stability, and predictive validity of between-person differences. By the early 1980s, this work had converged on the Big Five (Goldberg, 1981; McCrae & Costa, 1987), enabling cumulative science and large-scale longitudinal research. Trait constructs demonstrated substantial differential stability across time, raters, and the lifespan (e.g., Bleidorn et al., 2022; Damian et al., 2019; Fraley & Roberts, 2005; Roberts et al., 2006; Robins et al., 2001), establishing personality as a robust science of individual differences. However, in prioritizing parsimony and generalizability, this paradigm increasingly defined personality in terms of between-person differences rather than the organization within the person.

In parallel, a social-cognitive tradition emerged focused on the processes underlying behavior within individuals. These approaches redefined personality consistency as structured variability, formalized in conditional models (Mischel, 1973; J. C. Wright & Mischel, 1987), which conceptualized personality in probabilistic terms as patterns of behavior across situations.

A key limitation of early conditional approaches, however, is that they functioned largely as descriptive models, capturing regularities in behavior without specifying the processes that generate them. The Cognitive-Affective Processing System (CAPS; Mischel & Shoda, 1995) represents a major conceptual advance in addressing this limitation. Rather than focusing on objective features of situations, CAPS centers on their psychological meaning for the individual. These meanings activate cognitive-affective units (CAUs; including encodings, expectancies and beliefs, affect, goals and values, and competencies), which dynamically interact within an interconnected system.

Crucially, these units are not independent. Their interactions give rise to stable patterns of activation that produce behavior, which in turn feeds back into the situation by altering what features are available, salient, or consequential for the individual. In this way, CAPS provides one of the clearest articulations of personality as a system of interacting processes that generate both stability and variability across contexts.

Unfortunately, the result of this innovation was not a resolution to the person-population problem, but rather a division of labor between two largely isolated research programs. Trait approaches captured stable differences between individuals, while social-cognitive approaches captured structured variability within individuals. Yet the relationship between these levels remained unresolved. Rather than a unified science of personality, the field developed uneven and partially disconnected traditions. In the United States in particular, trait-based approaches

retained institutional dominance, whereas social-cognitive approaches, despite their focus on processes unfolding within persons, were often treated as peripheral to personality psychology, contributing to their relative absence from its central venues (Cervone, 2005; Cervone & Shoda, 1999; Lamiell, 2026; Mischel, 2004).

2.7 Outlook: Personality Science in the 21st Century

Arguably, the trait-based paradigm has become *the* science of personality, with traits treated as synonymous with personality both within and across fields (Rauthmann, 2024). Contemporary research has largely focused on how they change across time, what distal life outcomes they predict, and why (e.g., Beck & Jackson, 2022a; Bleidorn et al., 2022; Graham et al., 2020; Ozer & Benet-Martínez, 2006; Roberts et al., 2006; Soto, 2019).

At the same time, the reintroduction of psychological and behavioral fluctuations via the Experience Sampling Method (ESM) has renewed interest in situational influences and personality dynamics (e.g., Fleeson, 2001; Jayawickreme et al., 2021; Kuper et al., 2021). These advances have been enabled by technological developments that allow for intensive, multivariate time-series data collection at scale (Csikszentmihalyi & Larson, 1987).

Despite this progress, these data have mostly been used to extend, rather than challenge, the trait paradigm. In many cases, momentary experiences (i.e., personality states) are treated as manifestations of, and evidence for, the Big Five (e.g., Fleeson, 2001), and analyses are framed in terms of population-level regularities (e.g., consensual effects of situations on trait-relevant behavior; Sherman et al., 2015). Yet if traits are defined by stable behavioral regularities, and personality states are measured as momentary expressions of those traits, then the explanans and explanandum are not fully independent: trait scores and aggregated personality states risk redescribing the same regularities at different timescales. What remains unexplained is how

those regularities are generated, maintained, and changed within persons. Consequently, the study of personality dynamics has largely remained conceptually anchored to the very framework it might otherwise challenge.

Methodologically, dominant statistical approaches to ESM data, such as multilevel models, have been critical for separating within- and between-person effects (Raudenbush, 2002). However, these models remain rooted in linear assumptions, are typically applied to a limited set of variables, and generally presume that individuals instantiate the same basic process, differing primarily in degree rather than kind (Liu et al., 2024). Even with random effects, heterogeneity is usually modeled as deviation around a common population structure, making these approaches poorly aligned with the multivariate, nonlinear, reciprocal, and potentially person-specific processes that likely characterize psychological phenomena.

As a result, there remains a disconnect between the phenomena researchers seek to explain (i.e. whole persons) and the tools used to explain them (i.e. population-level approaches; Beck & Thapa, 2024; Long, Lewis, et al., 2026). Theoretical, methodological, and technological advances have made it possible to study the person as a dynamic system, yet these advances have primarily been used in ways that reinforce population-level constructs. In this sense, the field is well-positioned to move beyond the trait paradigm but has largely failed to do so. The question, then, is not only why the study of the whole person has re-emerged, but why it has not yet fulfilled its promise.

3 Why (Personality) Psychology Requires Person-Specific Inference

Against this backdrop, the continued dominance of the trait paradigm (Lamiell, 1986; Phan et al., 2025; Rauthmann, 2024) raises a more fundamental issue than is typically acknowledged: Can population-level methods and models built on between-person variation

adequately capture phenomena that reside within persons? More specifically, are individual differences approaches suited to the ontological level at which personality is constituted?

In what follows, we evaluate these approaches from conceptual, formal, epistemological, and inferential perspectives. Across these perspectives, a consistent pattern emerges: Linking population-level findings to processes unfolding within individuals over time requires additional assumptions that are rarely made explicit and often not satisfied.

3.1 Conceptual-logical Limitations of Population-level Approaches

Psychological phenomena, such as experiences, behaviors, and their regularities, are located within individuals, whereas individual differences exist only at the level of populations (Lamiell, 1981, 1997). They thus occupy fundamentally different conceptual levels.

Consequently, population-level approaches are inherently misaligned with the phenomena they aim to explain from a conceptual and logical standpoint.

Population-level statistics characterize distributions, such as central tendencies, variation, and covariation, across individuals but do not, on their own, justify inferences about processes within individuals. Even robust aggregate patterns do not imply that the same relations hold within individuals, as population-level regularities may coexist with substantial between-person heterogeneity (Drobisch, 1867; Lamiell, 2018).

Trait taxonomies illustrate this issue. Derived from patterns of covariation across individuals, they may reflect robust population-level structure but are often interpreted as causal entities within individuals. Such interpretations require strong assumptions – shared structure, comparability across persons, and primarily quantitative differences – that are rarely tested and often untenable (Borsboom et al., 2003).

Together, these considerations point to a mismatch between the epistemological level of population-level research and the ontological level of psychological phenomena. The question, then, is under what conditions inferences across these levels are warranted.

3.1.1 The Non-Ergodicity Problem

The ergodicity argument formalizes the conditions under which inferences from population-level statistics to within-person processes are justified (Molenaar, 2004; Molenaar & Campbell, 2009). Ergodicity concerns the equivalence between ensemble averages (across individuals) and time averages (within individuals), which requires two assumptions that are rarely tenable in psychological research: homogeneity and stationarity.

Homogeneity implies that the same underlying model applies to all individuals, such that they are interchangeable with respect to the process under investigation. This assumption conflicts with the existence of meaningful individual differences. Stationarity requires that the statistical properties of a process remain constant over time, which is at odds with path-dependent processes such as development, adaptation, and learning.

Violations of ergodicity have also been demonstrated empirically. For example, Borkenau and Ostendorf (1998) conducted person-specific principal components analyses on repeated personality assessments. While the population-level (R-technique) structure closely resembled the Big Five, person-specific (P-technique) structures diverged substantially in both the number of factors, loading patterns, and factor content, indicating that aggregate structures do not generalize to individuals. Similar findings have been reported by Molenaar (2004) and Beck and Jackson (2020a), showing that averaging over time obscures meaningful within-person variation.

Psychological processes are shaped by biological constraints (e.g., Kandler et al., 2021), environmental affordances (e.g., Roberts, 2018), and biographical histories (e.g., Bleidorn et al., 2018; Luhmann et al., 2012), making them heterogeneous, dynamic, and path-dependent. Under these conditions, homogeneity and stationarity are unlikely to hold. When either assumption is violated, the equivalence between population-level and individual-level patterns breaks down. Thus, even accurate population-level models do not, on their own, license inferences about within-person processes.

3.2 Epistemological Roots of Population-Level Approaches

If the conditions for valid person-level inference are rarely met, the question becomes why population-level approaches nevertheless dominate psychological research. From an epistemological perspective, their prominence is closely tied to “misconceived nomotheticism” (Uher et al., 2025) and the widespread adoption of the hypothetico-deductive model (Haig, 2014).

In psychology, the hypothetico-deductive logic is typically implemented by statistically testing predicted relations between variables across a sample of individuals. In practice, such approaches presuppose that phenomena follow general laws that hold uniformly across individuals, implicitly treating persons as interchangeable instances of a common underlying model (Danziger, 1990; Hamaker, 2012). Crucially, this deductive testing logic is distinct from the additional inductive step of generalizing from sample-level findings to populations.

Rather than reflecting an optimal match between method and phenomenon, the dominance of this approach can be traced to historical and institutional developments, including large-scale assessment practices and the influence of natural science models (Danziger, 1990). While well-suited to stable, homogeneous systems, they are poorly matched to complex, person-

level processes (Molenaar, 2004). As a result, research often rests on assumptions, such as lawfulness across individuals and interchangeability, that are rarely examined and often misaligned with the contextual, dynamic, and person-specific nature of psychological phenomena.

3.3 Person-Specific and Bottom-Up Approaches as an Epistemological Necessity

Taken together, these considerations suggest that population-level approaches are unlikely to provide an adequate basis for knowledge about persons. Their limitations are not merely methodological but reflect a deeper mismatch between dominant epistemological conventions and the ontological nature of psychological phenomena. This mismatch stems from the inherent complexity of such phenomena: They unfold dynamically within individuals (Cervone, 2005), who constitute higher-order complex systems embedded in and continuously interacting with their complex environments (Rauthmann, 2021). Accordingly, they cannot simply be isolated from the persons and environments in which they occur (Mischel & Shoda, 1995).

Consequently, psychological phenomena rarely conform to well-defined problems in which all relevant variables, relations, and boundary conditions can be specified. Instead, they are often underspecified with respect to measurement (Uher, 2025), formalization (Eronen & Romeijn, 2020; Meehl, 1990), and the auxiliary assumptions required to interpret pertinent empirical findings (Meehl, 1990; Peikert, 2023). In this sense, many psychological questions resemble ill-defined problems, where neither the relevant variables nor their interrelations are fully known in advance (Simon, 1973).

A more suitable alternative lies in person-specific, bottom-up approaches, which ground inference at the level of individuals – where psychological phenomena occur – and build

generalizations from these processes upward (Phan et al., 2025). Instead of presupposing fully specified hypotheses about general laws, such approaches proceed inductively; identifying regularities as they manifest within individuals and cautiously extending them across cases, rather than inferring them indirectly from interindividual variation. This enables the generation of knowledge that remains anchored at the person level while still allowing for more general insights across individuals and explicitly acknowledging the uncertainty of generalization. Unlike population-level approaches, these frameworks do not assume homogeneity but treat heterogeneity as a substantive feature of psychological phenomena.

In this way, bottom-up approaches offer an epistemological pathway that is better aligned with the ontological status of psychological phenomena as situated within individuals. We return to these approaches in greater detail in the implications and roadmap sections.

3.4 Implications for Inference and the Status of the Person

Beyond their conceptual and formal limitations, population-level approaches also create systematic problems in how empirical findings are interpreted and applied. They are well-suited to questions about groups or policies, particularly when the goal is to improve decisions and optimize outcomes across many people (e.g., in large-scale selection or targeting where even modest associations can shift aggregate outcomes). At the same time, they often provide only limited value for understanding or predicting outcomes in specific individuals (Modersitzki et al., 2025), and their relevance to the individual cases is frequently overestimated. This problem is not merely technical; it reflects a broader difficulty in reasoning about probabilistic information. As Möttus (2022) demonstrates, even correlations that are meaningful for population-level prediction often produce only modest gains in predicting individual people: Knowing that a person falls in the high range of a predictor typically increases, but does not determine, the

probability that they will fall in the corresponding high range of an outcome. For example, in our own individual participant data meta-analysis of personality and dementia, higher neuroticism and negative affect and lower conscientiousness, extraversion, and positive affect predicted dementia diagnosis across 44,531 participants (Beck et al., 2024). These associations are useful for characterizing population-level risk, but they do not imply that any particular person high in neuroticism or low in conscientiousness will develop dementia, nor do they explain the person-specific pathways through which risk unfolds. Population-level findings may therefore be useful at the aggregate level without providing commensurate knowledge about persons.

These limitations in person-level inference reflect a deeper difference in the status afforded to the person by different methodological orientations. Population-level approaches treat individuals as interchangeable units contributing to aggregate regularities, with explanatory emphasis placed on variables and their relations, whereas person-specific approaches treat the individual as the primary unit of analysis and seek to understand psychological phenomena as organized within the person as a whole (G. W. Allport, 1961). This asymmetry has ethical implications. Decisions guided by population-level evidence may be defensible in aggregate yet disadvantage individuals who are poorly represented by the underlying model (Fisher et al., 2018). From the standpoint of those affected, such regularities may offer little guidance for understanding, anticipating, or intervening on their own outcomes.

Person-specific approaches, by contrast, aim to generate knowledge that is directly applicable to individuals (Nissen & Beck, 2025b). Recognizing these limitations is therefore not simply a matter of technical refinement. If personality science aims to explain phenomena as they exist within persons, it must adopt frameworks capable of representing individuals as organized, dynamic, and contextually embedded systems.

4 A Primer on Systems

In Section 2, we tracked historical conceptualizations of personality, emphasizing tensions between the whole (person) and its parts (traits and affective, behavioral, cognitive, motivational, and situational factors). We argued that developments in the natural sciences, the rise of factor analytic tools, and the relative ease of collecting person $\times$ variable data contributed to the dominance of population-level traits. In Section 3, we further argued that this paradigm is inadequate for capturing both the wholeness of the person and its constituent processes.

At its core, this chapter asks: *What is personality?* Our claim is that existing paradigms are insufficient for answering this question. An adequate framework must integrate core features emphasized across definitions of personality while also addressing the conceptual and formal challenges outlined in prior sections (see Table 1).

Across definitions of personality, several key concepts are commonly reiterated including that personality is (1) “characteristic patterns” (Funder, 2010; Pervin et al., 2005), (2) “stable” but “changing” over (3) short (Epstein, 1980; Fleeson, 2001) and (4) long periods of time (Bleidorn et al., 2022; Caspi et al., 2005; Roberts et al., 2006) and (5) in reaction to environments (Kuper et al., 2024; Mischel & Shoda, 1995; Rauthmann, 2021), (6) coherent (sometimes referred to as “functional unity”; Allport, 1961; Cervone & Shoda, 1999; Fournier et al., 2015; Shoda et al., 1993), (7) multi-level (e.g., from molecules to culture; Benet-Martínez et al., 2015; Schwaba et al., 2025), (8) fundamentally a property of person (G. W. Allport, 1937; Beck & Jackson, 2020a; Stern, 1917), and (9) different across people (McAdams & Pals, 2006; Roberts & Yoon, 2022). Any adequate paradigm of personality must be capable of accounting for each of these features simultaneously.

Prior sections also identify additional constraints that any suitable framework must accommodate, including: (10) non-ergodicity and (11) the multi- and equifinality of psychological phenomena across levels, (12) a principled account of how levels are linked, and (13) the nonlinear dynamics that characterize complex systems.

In this section, we present an alternative: Conceptualizing personality as a person-specific complex, nonlinear dynamic system characterized by emergent properties, which we argue holds promise for addressing these criteria in a generative framework for the field. A dynamic systems account treats these questions as central rather than residual. To be clear, we do not propose a more dynamic trait model; we propose a model in which traits are not the primary mode of explanation. This framework can accommodate traits as descriptive summaries of system behavior but does not require them as foundational constructs; they are outcomes of the system rather than its sole organizing principle. After introducing core terminology and outlining dynamic perspectives on the single case, including supporting empirical evidence, we then turn to many-case dynamics via bottom-up integration. This ambitious proposal echoes the grand theories that marked the early days of personality theorizing. This is not accidental: In our view, personality psychology has overly focused on the parts over the last half century, missing the whole.

Table 1
Necessary Components of an Adequate Personality Paradigm

#	Description	Systems Feature
Criteria from Personality Definitions		
1.	Characteristic patterns of thinking, feeling, and behaving	Topological structure (i.e., stable regions in state space)
2.	Stability across longer periods of time	Attractor stability and persistence over time
3.	Change across longer periods of time	Topological change (e.g., emergence, disappearance, or transformation of attractors)
4.	Fluctuations across shorter periods of time	Multi-timescale dynamics and sensitivity to perturbations
5.	Interaction with the environment	Open system dynamics (person-situation coupling)
6.	Coherence in the internal patterning	Self-organization of multivariate structure
7.	Multilevel (molecules to cultures)	Emergent hierarchical organization across levels

8.	A (whole) person	Integrated, self-organizing system with emergent organization
9.	Differences across people	Inter-system heterogeneity in structure and dynamics (person-specific organization)
Criteria from Formal Scholarship and Empirical Work		
10.	Non-Ergodicity	Interindividual heterogeneity and intraindividual non-stationarity
11.	Multi- and equifinality of psychological and behavioral phenomena	Nonlinear interactions among components enabling multiple pathways to similar outcomes (equifinality) and divergence from similar starting points (multifinality)
12.	Account for the links among levels	Cross-level coupling via emergence and top-down constraint
13.	Non-linearity	Nonlinear dynamics (e.g., thresholds, asymmetries, and state-dependent change)

Note. Criteria are derived from personality definitions as well as formal and empirical constraints on psychological phenomena. Systems features describe properties required of any model that satisfies these criteria.

4.1 Introduction to Complex Systems Terminology

Dynamic and complex systems theories are closely related, differing primarily in emphasis: Dynamic systems theory focuses on how systems change over time (Katok & Hasselblatt, 1995), whereas complex systems theory focuses on patterns that emerge from interactions among components (Mitchell, 2009). Both are relevant to psychology (Guastello et al., 2009; Maas, 2024), which is concerned with processes unfolding across time and with phenomena that are multiply determined (Molenaar, 2004).

At the same time, systems terminology is often used imprecisely, contributing to conceptual ambiguity (e.g., Brown et al., 2013). The goal of this section is therefore to provide a focused introduction to core concepts that will be used throughout the remainder of the chapter. Rather than defining each term repeatedly in the text, we provide working definitions and psychological examples for the core systems concepts in Table 2. To make these concepts more concrete, consider a person's sociability across time in everyday life. On some days, they readily seek out interaction; on others, they withdraw. These shifts may reflect external demands, such as work meetings or invitations from friends, as well as internal states, such as fatigue, mood,

goals, or expectations about how an interaction will go. A dynamic systems framework asks not only whether this person is generally more extraverted or introverted, but how internal and external forces combine over time to make some patterns of sociability stable, others more fragile, and still others likely to change under particular conditions.

Several features of systems are particularly relevant for psychological phenomena, like sociability. First, many psychological processes are **non-stationary**, exhibiting changes in means and variability over time (Fisher et al., 2018; C. J. Lee & Beck, 2025a). For example, many individuals experience higher levels of negative affect on working days relative to weekends and holidays (Ryan et al., 2010; Stone et al., 2012), resulting in a stark mean shift in a time series. Longer term, phenomena like seasonal affective disorder (SAD) result in clear seasonal shifts in depressive symptoms during winter months relative to the rest of the year (Rosenthal et al., 1984). While such non-stationarity poses challenges for common statistical models, it is readily accommodated within systems frameworks.

Second, psychological phenomena often exhibit wholeness: Personality is more than the sum of its parts (G. W. Allport, 1937; Beck & Jackson, 2020a; Cramer et al., 2012; McAdams & Pals, 2006; Stern, 1917). This is characteristic of complex, nonlinear systems in which **self-organization** and **emergence** can produce higher-order structure from lower-level interactions (Artibe & De Domenico, 2022). Self-organization refers to the process through which order arises from interactions among system components rather than being imposed by a central controller (Nicolis & Prigogine, 1977), while emergence refers to the appearance of higher-level patterns that are not reducible to individual components (Hoel et al., 2013). Critically, this still allows for top-down effects, often conceived of as system-level constraints, but does not imply the presence of a top-down “controller” (Juarrero, 1999, 2023).

Third, psychological phenomena are often shaped by both internal (e.g., cognitive, biological) and external (e.g., situational, cultural) factors (Beck & Jackson, 2022c; Kuper et al., 2024). In systems terms, these correspond respectively to endogenous and exogenous regulatory forces that govern the self-organization of the system (Collier, 2004). In practice, these distinctions are analytic rather than absolute: internal goals, needs, and self-regulatory processes are themselves shaped by social contexts, while external norms and affordances matter only insofar as they are perceived and acted upon by persons. For example, self-regulation (Hoyle, 2006; McCrae & Löckenhoff, 2010) and goals (Heller et al., 2007; McCabe & Fleeson, 2016; Read et al., 2010) can be understood as endogenous forces, whereas social norms (Bicchieri et al., 2023; Mischel & Shoda, 1995) and environmental affordances (Rauthmann et al., 2015; Rauthmann & Sherman, 2020) are exogenous. Even the constant, heterogeneous sensory stream each person encounters requires some degree of organization (e.g., Grossberg, 1982). Without it, psychological processes would devolve into chaotic responses to minor fluctuations in input.

Fourth, most psychological systems are fundamentally **open systems** because they are in constant interaction with other systems (e.g., a person in interaction with other persons, the situations they occupy, or culture; Mischel & Shoda, 1995; Rauthmann, 2021; Sosnowska et al., 2019). Yet, common traits may be understood as quasi-**closed systems**, insofar as their state is assumed to remain constant rather than vary as a function of other systems (G. W. Allport, 1960). This implies that traits should not fundamentally shift following **perturbations** (Caspi & Moffitt, 1993). At the same time, contemporary accounts often invoke bottom-up processes to explain trait change (Jackson & Wright, 2024; Wrzus & Roberts, 2017), creating a tension between assumptions of short-term stability and mechanisms of longer-term change. Notably, this tension persists even when traits are conceptualized as emergent from lower-level dynamics,

as different levels of a hierarchical system may exhibit fundamentally different properties and system dynamics (Anderson, 1972).

Fifth, many psychological phenomena are characterized by both stability and change (Caspi et al., 2005). This duality is largely incompatible with the traditional factor models in which dispersion around the stable components is considered error variance (Borsboom et al., 2003). Yet in dynamic systems, stable aspects are often described as **equilibria**, **steady states**, or **homeostasis**, depending on system properties (Keller, 2008). Because we are concerned with the person as an open system, our focus is on homeostasis (Ashby, 1947; Keller, 2008, 2009; Maturana & Varela, 1980; see Table 2 for distinctions). Systems are often described in **topological** terms (Butner et al., 2015) – that is, in terms of the qualitative structure or “shape” of the system’s state space – which help to characterize different forms of homeostasis. In psychology, an **attractor** is a particularly important topological feature (Heino et al., 2023; Nowak et al., 2005), often illustrated as a basin toward which the system tends to evolve following perturbation (Thelen & Smith, 1994). More complex topologies, including multiple attractors, are also possible (Büchner et al., 2025; Kugler et al., 1991). The strength of topological features are often used as summaries of the strength of **self-organization** in the system (Ashby, 1947; Butner et al., 2015; Nowak et al., 2005).

Table 2
Definitions of Core Systems Concepts

Concept	Definition	Example
Basic Terminology		
State	The condition of a system at a particular moment or time, as described by all of the parameters that contain information about that system (Newman et al., 2019).	State effects are presumed to taint tendency measures by capturing non-relevant elements of a person's psychological experience in that moment (e.g., a "bad day"; Schwarz & Clore, 1983).
State Space	A topological representation of all possible states of a dynamical system. (Butner et al., 2015).	The range of possible configurations of a person's thoughts, feelings, and behaviors (e.g., the full space of momentary experiences).
Topology / Topological Structure	The qualitative shape or organization of a system's state space, including regions toward which the system tends to move, boundaries between regions, and pathways by which states change over time.	The "landscape" of a person's psychological states, such as whether their experiences tend to return to one stable pattern, shift between two distinct patterns, or remain highly variable across situations.
Perturbation	A temporary or sustained disturbance, change, or fluctuation that deviates a system from its normal state, equilibrium, or predicted behavior.	Receiving negative feedback, experiencing a social rejection, or encountering an unexpected stressor that shifts mood or behavior.
Closed System	A system that does not allow transfer of energy, information, or matter inside or out of the system. Closed systems cannot be self-organizing because they proceed toward increasing entropy, making the spontaneous generation of order unlikely.	Personality traits conceptualized as fixed, context-independent dispositions.
Open System	A system that allows the transfer of energy, information, or matter inside or out of the system. Open systems may be characterized by self-organization.	A person's emotions or behavior changing in response to social interactions, environmental demands, or cultural norms.
Non-stationarity	A property of any stochastic process whose moments are different if sampled at different points in the process. More broadly, and as used in this chapter, any process with non-constant mean and variance across time, either within one person or across persons (Newman et al., 2019).	Changes in emotion variability across a stressful semester or major life transition.
Ergodic fallacy	The fallacious assumption that the average behavior of a group applies to individuals across time. This assumption is only appropriate when individuals are so similar they are interchangeable and individuals exhibit stationarity over time.	Assuming that the average association between stress and emotion across people applies to how stress affects any single individual.
Homeostasis		
Equilibria	The point in a closed system in which all opposing forces are balanced, meaning there is no net change.	A hypothetical fixed level of a trait assumed to remain constant regardless of context.
Steady State	The point in an open system in which the inputs and outputs of the system are balanced, allowing the internal conditions of the system to remain constant; often considered an energy burning system.	A stable level of daily mood maintained despite ongoing fluctuations in daily experiences.
Homeostasis	The process of maintaining stable internal conditions, despite changing external conditions.	Returning to a typical mood or level of well-being after temporary stress or emotional disruption.

Attractor	A state or set of states toward which a system tends to evolve.	Recurring patterns of emotional or behavioral states toward which a person's experiences tend to return over time, such as a person repeatedly returning to a low-anxiety state after stress.
Multistability	The property of a system to have two or more coexisting stable states (e.g., attractors), allowing it to switch between distinct, mutually exclusive modes of functioning.	A tendency to express two distinct patterns across time, such as someone who tends to settle into either highly sociable or highly unsociable modes.
Regulatory Forces	Endogenous and exogenous influences that shape the direction and stability of system dynamics, governing how the system responds to perturbations and maintains or shifts its organization.	Goals, affective tendencies, and environmental demands jointly shaping whether a person returns to a typical pattern of behavior or transitions to a different one.
Emergence		
Self-Organization	A process in which spontaneous order arises from a heterogeneous system, in which elements of a system interact in a way that increases their structure or complexity (Newman et al., 2019).	Stable patterns of behavior (e.g., procrastination habits or social engagement patterns) emerging from repeated interactions among goals, emotions, and environmental demands.
Emergence	New patterns, processes, or structures that appear at higher levels of organization in the observation of phenomena that are not present at lower levels of organization (Newman et al., 2019).	The emergence of trait-like regularities from the repeated interaction of cognitive, affective, behavioral, and situational processes. "Being conscientious" emerging from routines, goals, feedback, affordances, and self-regulation rather than a single inner cause
Feedback Loops	Reciprocal interactions in which components of a system influence one another over time, such that outputs of the system feed back into future inputs, enabling the amplification or stabilization of patterns.	Repeated interactions between mood, behavior, and environment (e.g., social withdrawal leading to fewer positive interactions, reinforcing low mood).

Personality is a Person-Specific Complex, Non-Linear Dynamic System

“Personality is the dynamic organization within the individual of those psychophysical systems that determine his unique adjustments to his environment” (Allport, 1937, p. 48).

Now that we have introduced core terminology, we turn to dynamic and complex systems theories as providing theoretical and empirical tools that address key deficiencies in current paradigms, integrate diverse perspectives, and adjudicate disagreements across approaches. In doing so, we argue that a dynamic systems perspective offers a generative map for the future of personality psychology (see Table 3), and, more broadly, for psychology writ large.

Table 3
Overview of Dynamic Theories

Theory	Brief Description	Relation to Present Framework
Cognitive-Affective Personality System (CAPS; Mischel & Shoda, 1995)	Explains patterned behavior through interactions among cognitive-affective units such as encodings, expectancies, affects, goals, values, and competencies, emphasizing if-then contingencies and psychologically construed situations	Closely aligned with the present framework in treating personality as an interacting process system that generates both stability across time and variability across contexts. However, CAPS stops short of fully theorizing the person as an emergent, multi-level dynamic system whose broader organization changes across longer timescales.
Cybernetic Big Five Theory (CB5T; DeYoung, 2015)	Posits Big Five traits as individual differences in the parameters of a cybernetic self-regulating system. Personality reflects how people pursue goals, monitor discrepancies between current and desired states, respond to feedback, and adjust thought, emotion, and behavior over time. Traits are not merely descriptive summaries of behavior, but dispositional differences in the mechanisms that govern goal-directed action, feedback sensitivity, and self-regulatory control.	Incorporates feedback and regulatory concepts; however, CB5T still retains the Big Five as central higher-order organizing constructs, whereas the present framework does not presuppose any fixed trait taxonomy as primary and instead begins with person-specific dynamics from which broader regularities may emerge.
Functionalist Approach (D. Wood et al., 2015)	Explains behaviors in terms of their functionality to the individual (i.e., as means toward desired ends). Behaviors are shaped by process variables such as self-efficacies/abilities, expectancies, values, and goals. Rather than covariation as a consequence of latent factors, covariation between two behaviors is observed when the same process variable(s) support both behaviors simultaneously.	Compatible with the present framework in emphasizing regulatory forces and shifting explanation away from broad trait factors and toward process variables. The present framework extends this by embedding functional relations between psychological behaviors and process variables within a broader systems framework that also addresses temporal regularities, coherence, and their person-specific nature.
Knowledge-and-Appraisal Personality Architecture (KAPA; Cervone, 2005)	Conceptualizes personality in terms of enduring knowledge structures and contextually-sensitive, dynamic cognitive-affective appraisal processes. Emphasizes the differentiation among beliefs, goals, and evaluative standards as distinct forms of mental content.	Converges with the present framework in foregrounding process, appraisal, and dynamic state construction. KAPA is less explicit, however, about emergence across levels and the whole person as a self-organizing system embedded in open person-environment transactions.
Network Perspective on Personality (Cramer et al., 2012)	Conceptualizes personality as a network of connected cognitive, affective, and behavioral components whose relations may be causal, homeostatic, or logical; traits are treated as descriptors emerging from component connectivity.	Explicitly treats trait-like regularities as emergent rather than foundational. The present framework is broader in scope, adding open systems, multi-level emergence, and a more explicit account of how organization may differ across persons and timescales.
Nonlinear Interaction of Person and Situation model (NIPS; Blum et al., 2018)	Person $\times$ situation model proposing that behavior follows a nonlinear, symmetric interaction between trait level and situational affordance. It argues that both situations and persons can be weak or strong: variability is greatest in the middle of the continuum and reduced at the extremes.	Compatible with the present framework in emphasizing nonlinearity and the joint shaping of behavior by person and situation. However, NIPS is much narrower in scope: it models the general shape of person-situation interactions for specific, consensual traits and behaviors, and does not address how personality is person-specific, multivariate, regulated, and dynamic.

PersDyn Model (Sosnowska et al., 2019)	Models personality states as a (univariate) dynamic system that fluctuates over time around an attractor, or a characteristic state toward which the system tends to return. Momentary deviations from that attractor capture short-term variability, and attractor strength reflects how quickly the system returns following those deviations. This return tendency is conceptualized as a form of personality regulation.	Closely aligned in treating personality states dynamically and in using attractor concepts. The present framework builds on this logic but treats attractor models as only one part of a broader framework that also includes multivariate dynamics, external regulatory constraints, and the possibility of multistable topologies (i.e., more than one attractor).
Person–Environment Relations Model (PERM; Rauthmann, 2021)	Systematizes interactions, correlations, fits, and transactions between person and environment variables across short- and long-term timescales. PERM frames person–environment relations not as a single process, but as a set of reciprocal processes that unfold over time and jointly account for stability and change.	Compatible with the present framework’s argument that personality is an open system and cannot be understood independent from ongoing transactions with environments. The present framework extends PERM by asking how such transactions help generate the topology, coherence, and higher-order organization of the person-system itself.
Three-level model / cues–tendency–action (CTA) framework (Revelle & Condon, 2015)	Dynamic model spanning within-person, between-person, and between-group levels. Traits are conceptualized as rates of change in states in response to environmental cues, with action tendencies exhibiting inertia and consummation (i.e., fulfillment), eventually accumulating into observable patterns of frequency and duration over time.	Strong precursor to the present framework as it treats personality dynamically, foregrounds within-person temporal processes, and acknowledges that structure often differs across levels. Still, CTA remains primarily a formal dynamic model of traits, states, and choices across levels of analysis, rather than a general account of personality as a broader multivariate system characterized by reciprocal coupling among processes, nonlinear change, and reconfiguration of system structure across time and context.
Triggering situations, Expectancy, States/State expressions, and Reactions Framework (TESSERA; Wrzus & Roberts, 2017)	Explains long-term personality development as arising from repeated short-term sequences in which triggering situations evoke expectancies, shape momentary states, and elicit internal or external reactions. These recurring sequences accumulate and consolidate into lasting change.	Shares the present framework’s emphasis on linking short-term dynamics to long-term development. TESSERA is inclusive in its constructs of interest, which include traits, motivational orientations, and life narratives. However, it primarily explains development as the accumulation of repeated sequence patterns, whereas the present framework treats personality as a system whose development may involve qualitative, and possibly abrupt, reorganization in system structure or topology.
Whole Trait Theory (Fleeson & Jayawickreme, 2015, 2025)	Conceptualizes personality as constituted by Big Five traits that include both descriptive and explanatory components. The descriptive component is represented by density distributions of personality states, whereas the explanatory component consists of underlying social-cognitive mechanisms linked within a spreading activation network.	Shares the goal of linking dispositional traits to momentary processes, but remains anchored to trait taxonomies as organizing constructs. The present framework goes further by treating trait-like regularities as possible higher-order summaries of system dynamics rather than as the primary unit around which explanation is organized.
Person-specific complex, nonlinear	Conceptualizes personality as a person-specific, complex, nonlinear, open, multi-level dynamic system. Characteristic	Distinctive features of the present framework are its commitment to person-specific inference, non-ergodicity, open systems,

**dynamic systems
framework (present
chapter)**

patterns are understood as emergent phenomena of ongoing interactions among cognitive, affective, behavioral, motivational, biological, interpersonal, situational, and cultural processes. No single consensual construct, including traits, is assumed to be foundational. Instead, higher-order regularities are treated as outcomes of self-organization that may differ in topology, timescale, and organization across persons.

bottom-up emergence across levels, and top-down constraints. It aims to provide a generative framework within which existing dynamic approaches can be organized, extended, and revised.

Note. The selection of approaches in this table was informed in part by Kuper et al.'s (2021) review of personality dynamics, but the present table is an original synthesis tailored to the present chapter's dynamic systems framework and comparative aims. With the exception of the present framework, approaches are alphabetically ordered.

4.2 Characteristic Patterns of Thinking, Feeling, and Behaving

*“To serve as a successful *modus vivendi*, a trait must remain plastic or lose its usefulness”*
(Allport, 1937, p. 332).

Perhaps the most consistent feature of personality definitions is that personality is characteristic patterns of thinking, feeling, and behaving. Yet what is a characteristic pattern? At minimum, it must capture tensions between long-term stability and short-term fluctuations (i.e., cross-situational (in)consistency). A characteristic pattern implies something that should be stable across longer periods of time (i.e., not only displayed for a short period), which is compatible with the idea of personality as a *disposition*². On another, a characteristic pattern may also imply a distinctive pattern of behavior across situations, which is compatible with personality as *reactivity* (or rate of change; Revelle & Condon, 2015) to internal and external input. While dispositions are the core of trait approaches, reactivities are more core to social-cognitive approaches.

Prior attempts to integrate stability and variability have relied on statistical summaries that either discard information or fail to capture the underlying dynamics of the system. For example, a growing body of work has examined whether within-person variability in positive and negative affect is associated with a variety of outcomes, such as psychological well-being (Dejonckheere et al., 2019; Hehlmann et al., 2025; Houben et al., 2015; Long, Reitz, et al., 2026; Long & Beck, 2026b). Having more variable negative affect, for example, may be characteristic of an unstable emotional system (i.e., strong fluctuations across time), leading to poorer well-being overall. Yet there is limited consistent evidence of even basic associations between such simple univariate dynamic metrics (Dejonckheere et al., 2019), leading many to write off the

² Notably, this framing is also compatible with models of personality that move beyond dispositions, such as McAdams’s (McAdams & Pals, 2006) three-level model in which narrative identity is assumed to show similar (if not stronger) stability to dispositional measures across time.

utility of the study of dynamics relative to simple averages across time. Yet, such approaches based on aggregated or univariate indicators of variability, among others, treat fluctuations as properties of isolated variables and cannot represent how processes interact over time. As a result, they are poorly suited for modeling multivariate, nonlinear, and person-specific dynamics (i.e., they assume linear additivity; Long, Lewis, et al., 2026).

Dynamic systems models address these limitations by capturing both dispositions and reactivities jointly using differential equations. Differential equations are rooted in the language of change (Batra, Beck, et al., 2025) because they aim to predict *changes* in a process based on the current state of the system. This is critical because it allows both stability (where the system tends to be) and change (how it responds to perturbation) to be modeled within the same framework. In this sense, dispositions and reactivity are not competing explanations, but jointly estimated properties of the same underlying system. In plain terms, the model asks: When a person is above or below their usual level on some process, *what happens next?* Do they remain there, drift farther away, or return toward a “characteristic” level? We leverage a simplified linear, univariate differential equation to draw on basic regression-based knowledge (for extended tutorials, see Batra, Beck, et al., 2025; Chen et al., 2025) in Eq. 1.

$$\frac{dx}{dt_{it}} = b_{0i} + b_{1i} * x_{it} + \epsilon_{it} \quad (1)$$

In regression terms, x is a single process measured for person i at time t , and $\frac{dx}{dt_{it}}$ is its momentary change for time step dt . b_{0i} captures expected change when $x=0$, and b_{1i} captures the association between the current state of the system x_{it} and its momentary change. In other words, we are predicting how a process changes based on its current level.

In systems terms, this is a simple attractor model capturing **homeostasis**, or the point at which the forces in the system are fully balanced such that the system will remain stable if not

for perturbations. Two components define this system: The **location** of the homeostatic point (the level toward which the system tends) and its **strength** (how strongly the system returns following perturbation). The homeostatic location is the point at which the system is not changing ($\frac{dx}{dt} = 0$), which yields:

$$\begin{aligned} 0 &= b_{0i} + b_{1i} * x_{it} \\ -\frac{b_{0i}}{b_{1i}} &= x_{it} \end{aligned} \quad (2)$$

In this simple model, attractor strength is captured by b_{1i} . A negative b_{1i} indicates that the homeostasis point is an **attractor**.

Figure 2 visualizes an attractor in three complementary ways. The left panels illustrate a strong attractor (i.e., the system returns more quickly to its homeostasis), whereas the right panels illustrate a weak attractor. In each panel, the x-axis represents the current level of x.

The top panels show direction of change: Values below the attractor increase, whereas values above decrease. The middle panels depict change as a function of state ($\frac{dx}{dt}$), where the zero-crossing identifies the attractor and the slope reflects its strength. The bottom panels represent the same system as a basin, where the steepness of the landscape reflects how strongly the system returns to homeostasis. Strong attractors (left) return quickly following perturbation, whereas weak attractors (right) return more slowly. The basin representation, in particular, provides an intuitive depiction of an attractor. Imagine that the open white circle is a ball dropped into the basin, and then we observe how it moves throughout the basin. The ball in the left basin (strong attractor) will move faster and can be expected to reach the homeostatic point (the bottom of the basin) more quickly than the ball in the right basin (weak attractor).

Psychologically, an attractor can be understood as a pattern the person tends to return to. For example, one person may usually return to a calm, engaged mode after a stressful interaction,

whereas another may remain irritable or withdrawn for hours. The point is not simply that the two people differ in *average* irritability or sociability, but that their systems differ in *how strongly* they return to particular states following perturbation.

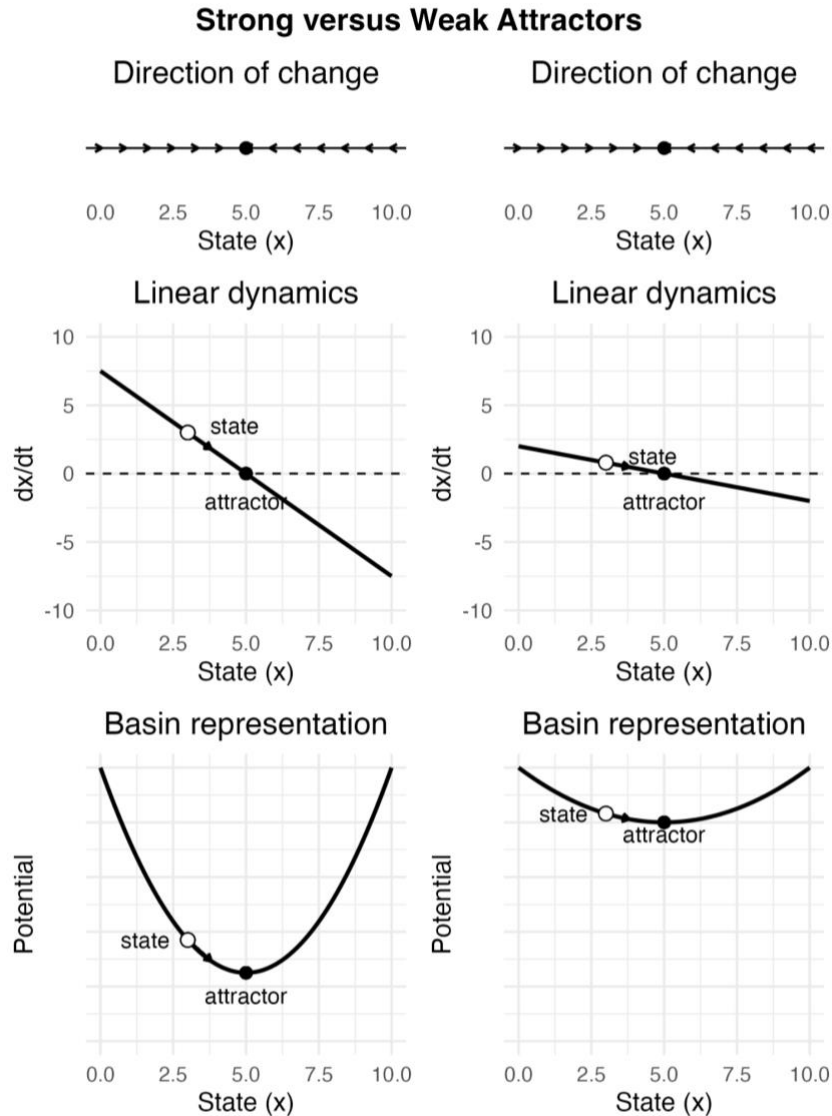


Figure 2. The same underlying system is visualized in three equivalent ways: direction of change across the state space (top), the state-dependent rate of change (dx/dt ; middle), and the corresponding potential (basin) representation (bottom). Together, these panels illustrate how attractor strength is a property of the system's organizing dynamics. Strong attractors (left) are characterized by steeper local dynamics and deeper, more curved basins, reflecting stronger regulatory forces and faster return to homeostasis following perturbation. Weak attractors (right) exhibit shallower dynamics and flatter basins, reflecting weaker regulatory forces and slower return. The white point denotes the current state, the black point denotes the attractor, and arrows indicate the direction of change implied by the system's dynamics.

More complex extensions of this model (e.g., nonlinear and multi-attractor systems) exist, but here we focus on linking these dynamics to dispositions and reactivity. Dispositions are captured by attractor location (where the system tends to be), whereas reactivity is captured by attractor strength (how the system responds to perturbation).

To evaluate whether these parameters capture dispositional tendencies and reactivity, recent work has examined their reliability, validity, and predictive utility. For example, Danvers and colleagues (2020) modeled Big Five personality states as univariate systems. Most individuals' states were well-described by a linear model with a single attractor. Attractor location was strongly associated with person-means ($rs \approx .95$), consistent with a dispositional interpretation, whereas attractor strength showed only modest associations with within-person variability ($rs \approx -.30$), indicating that it captures a distinct aspect of system dynamics.

These models were estimated using approximations (i.e., time-normalized difference scores) that may introduce bias and obscure underlying dynamics (Büchner et al., 2026). More broadly, applying dynamic systems models to psychological data has historically required such approximations. However, recent methodological advances (Driver et al., 2017) now allow for more direct estimation, enabling closer alignment between theoretical models and empirical tests.

Using updated dynamic systems tools, we examined the predictive validity of attractor location (tendency) and strength (reactivity) of personality states for distal outcomes (e.g., depression, life satisfaction) over two years (Long & Beck, 2026b). We also compared their incremental validity relative to “aggregate” theories (intraindividual mean (iMean) and intraindividual standard deviation (iSD), respectively) as well as global Big Five ratings. Two findings have particularly strong implications for personality theory.

First, person-means ($r \approx .26$) and global trait measures ($r \approx .20$) performed similarly in magnitude but person means ($\approx 60\%$) predicted a larger proportion of outcomes than global traits ($\approx 30\%$), suggesting that aggregate momentary experiences are more closely tied to life outcomes than global self-reports, which may more strongly reflect self-concept. Given only moderate convergence between trait and state measures ($r \approx .40$; Fleeson & Gallagher, 2009) and theories positing that mean states directly correspond to traits (Fleeson & Jayawickreme, 2015, 2025), this points to a predictive cost of relying on global measures over ecologically contextualized assessments.

Second, we observed consistent interactions between location and strength in predicting outcomes, indicating that *jointly* considering them improves predictive utility. For example, individuals with both higher average levels and greater stability of momentary depression show worse well-being than those elevated on only one dimension. Conceptually, this supports the view that personality reflects both general tendencies (dispositions) and how individuals enact them (reactivity).

In another study, we linked dispositions and reactivity to a foundational concept in personality theory: Gordon Allport's central traits (Long & Beck, 2026a). As previously noted, Allport argued that most individuals can be described in terms of a distinct set of central traits around which the rest of personality is organized³. Critically, this implies that which personality states are central should vary across persons, as should their organization within the system. In this project, we examined attractor location and strength as two potential operationalizations of trait centrality – central traits as especially salient *tendencies* (higher attractor locations) and

³ Although the term “cardinal traits” has gained more recognition, Allport believed cardinal traits to be rare, present only individuals for whom a single dominant tendency could be identified (Allport, 1937, pp. 338). Thus, this line of work has focused on individuals' likely more diversified set of central traits, which Allport believed to be more common.

central traits as especially *reactive* patterns of behavior (higher attractor strength). Across five samples, we (1) quantified heterogeneity in attractor strength and location within and between individuals and (2) tested whether these parameters mapped onto participants' open-ended central trait descriptions.

We observed substantial heterogeneity in both parameters across persons and states, consistent with the idea that personality is *both* tendencies and reactivities. However, contrary to our expectations, attractor strength and location were not related to subjective perceptions of trait centrality. This null finding highlights a broader tension in integrating modern and historical, as well as qualitative and quantitative, approaches. Mapping open-ended central traits onto state-level measures required imposing Big Five structure, which may be incommensurate with non-ergodicity.

In sum, the question of what is characteristic in personality is foundational, particularly given divergent perspectives on whether patterns reflect tendencies (dispositions) or discriminative responses (reactivity). From a dynamic systems perspective, this tension is, in part, an artifact of the framework itself: Personality is both what people tend to do (attractor location) and how they respond to perturbation (attractor strength). These are not competing explanations, but complementary features of the same system. However, personality is more than a set of variables; it is about understanding the whole person. We turn to this next.

4.3 The Whole Person

“Instead of growing impatient with the single case and hastening on to generalization, why should we not grow impatient with our generalizations and hasten to the internal pattern?”
(Allport, 1961, p. 84).

Why is the whole person the fundamental starting point for understanding personality? First, psychological phenomena unfold within the person, and persons are inherently wholes (i.e.,

Stern's [1916] *unitas multiplex*). Emotional *experience*, for example, occurs within individuals and only becomes part of the interpersonal world through emotional *expression* (Gross et al., 2000; Scherer, 2005). Consequently, to understand emotions across people, we must first understand how they operate within a person.

Second, typical approaches assume a common cause model in which latent traits causally generate observed indicators (Borsboom et al., 2003). Under this assumption, if Agreeableness causes politeness, then the cause of politeness should be the same across individuals. Yet, people behave politely for different reasons (i.e. demonstrating equifinality), including a desire for power, a favorable view of others, or adherence to social convention (Baumert et al., 2017). This heterogeneity implies that indicators are not exchangeable, violating a core assumption of common cause models.

Third, the lack of exchangeability implies that personality cannot be reduced to a single underlying cause. If the same behavior can arise from different processes across individuals, it must reflect the joint influence of multiple processes operating within a person. Consequently, personality is inherently multivariate. In one study, we investigated personalized ($N=1$) machine learning models of six momentary behaviors and experiences (e.g., loneliness, procrastination) using ESM data (Beck & Jackson, 2022c). For each person and outcome, we compared model performance across three feature sets: Psychological features, situational features, and their combination. We found that (1) behaviors could be predicted with reasonable accuracy, (2) individuals differed in whether psychological, situational, or both features best predicted their behavior, and (3) the specific predictors that mattered varied substantially across individuals. We also replicated these results in the realm of education and student performance (Saqr et al.,

2024). Together, these findings indicate that behavior and experience are multiply determined in heterogeneous ways across individuals.

Fourth, research on the multivariate structure of personality within individuals (i.e., Raymond Cattell's P-technique) shows that person-specific structures do not map onto population-level structures (i.e., R-technique; Beck & Jackson, 2020a; Borkenau & Ostendorf, 1998; Molenaar, 2004; Shimotsukasa & Mieda, 2026). Using both P-technique factor analysis and time-series network approaches, we found that person-specific personality structures were reliable and relatively consistent across time (mean profile $r_s \approx .62$, comparable to the ipsative stability of Big Five traits over similar intervals; Robins et al., 2001), yet did not resemble population-level structures, indicating non-ergodicity between levels (Beck & Jackson, 2020a). These findings have been replicated across the COVID-19 pandemic (Beck & Jackson, 2021b), across one year (Beck & Jackson, 2022b), using time-varying network approaches (C. J. Lee & Beck, 2025b), and in a sample of Japanese undergraduates (Shimotsukasa & Mieda, 2025). Moreover, individual differences in person-specific structures are associated with personality trait change and concurrent and distal life outcomes (Nissen et al., under review; Nissen & Beck, 2025a).

This whole person approach can be extended by examining not only average structure but also recurring momentary organizations. That is, individuals may exhibit multiple, recurring profiles of personality expression over time. Using Cattell's O-technique, Lee and Beck (2025a) identified idiographic momentary profiles: Person-specific clusters of time points defined by similar configurations of personality facets. Across two samples, the majority of participants (66.4%, 80.6%) exhibited more than one profile of personality expression. Personality is thus characterized not only by stable person-specific structure, but also by patterned variability in

how that structure is expressed across time. From a systems perspective, such recurring profiles can be understood as distinct regions of a person's dynamic landscape that may indicate the presence of multiple homeostatic points, also known as multistability (see Table 2).

In a complex system, non-ergodicity and non-exchangeability are not failures of the data-generating process, but signatures of a weakly emergent system in which consistent phenomena arise from reciprocal, context-dependent interactions among components and the environment (Chalmers, 2006). Consequently, complex systems lack a one-to-one mapping between the whole (e.g., personality traits) and its parts (e.g., specific behaviors). Therefore, psychology must prioritize understanding phenomena within a single person, even when the goal is to explain differences across people (Nissen & Beck, 2025b).

In this section, we argued that personality is fundamentally a property of a multivariate, whole person. Yet what this “wholeness” – often described in terms such as functional unity (G. W. Allport, 1937, 1961) or coherence (G. W. Allport, 1937; Caspi & Moffitt, 1993; Cervone & Shoda, 1999; Fournier et al., 2015) – entails has long challenged psychology (Lamiell, 2026; Murray, 1938). We suggest that self-organization, or the process through which heterogeneous interactions give rise to emergent patterns, provides a framework for understanding this coherence and reframes wholeness as a dynamic pattern, not a static one. We turn to this next.

4.4 Personality Coherence and Self-Organization

“Let me repeat this question, for it is the one that more than any other has haunted me over the years. What makes the system cohere in any one person?” (Allport, 1960; p. 308).

If behavior and experiences are equifinal, multifinal, and multicausal, then how do psychological regularities emerge at all, particularly when small perturbations can have outsized consequences. Trait theories answer this by positing internal, mechanistic causes (e.g., by tying traits to universal components of cognitive processes; DeYoung, 2015), or to biologically

embedded basic tendencies (McCrae & Sutin, 2018). Yet this explanation is circular: Traits are both summaries of behavioral regularities and invoked as their source. In other words, the phenomena to be explained (behavioral regularities) are built into the definition of the explanatory constructs (traits).

This explanatory problem invites two common responses. The first is to treat traits as descriptive summaries of characteristic patterns of thinking, feeling, and behaving that are not meant to explain how those patterns are generated (F. H. Allport, 1937; Buss & Craik, 1983). This response avoids circularity by limiting the explanatory role of traits. From this perspective, traits may efficiently describe behavioral regularities, support prediction, organize individual differences, and provide a shared language for comparing people. However, if traits are descriptive summaries, they cannot also serve as explanations of the same regularities they summarize. Thus, a descriptive trait model may tell us that a person tends to be polite, sociable, anxious, or persistent, but it does not explain how those tendencies are produced, why they vary across contexts, or why they remain stable or change over time. Descriptive trait models remain valuable, but their value should not be mistaken for an explanation of personality expression.

The second response is to treat traits as latent propensities that generate characteristic patterns of expression. This response preserves the explanatory ambitions of trait theory, but only by taking on a different burden. Consider again the example of politeness. A trait account might hold that a person behaves politely because they possess a higher latent propensity for Agreeableness. Yet if Agreeableness is meant to explain polite behavior, then a theory must specify how this propensity operates within the person, which situations invoke it, how it interacts with those situations, and why it produces politeness in some contexts but not others. It must also explain why people with similar levels of a latent propensity may express it differently.

Without such an account, the latent propensity risks simply redescribing the regularity rather than explaining the organization that generates it.

A dynamic systems perspective begins from this explanatory gap between description and explanation of behavioral regularities. Rather than asking whether traits are best understood as summaries or propensities, it asks how stable patterns emerge in complex, person-specific systems. The answer is **self-organization**: Order emerges from interactions among system components, without a central organizing cause (e.g., Ashby, 1947; Keller, 2008; see Table 2). On this view, traits, consistency, and stability are not primitive causes but emergent outcomes of person-specific dynamics. Traits are real in the way climate is real: Not a single mechanism, but a stable pattern that organizes and constrains lower-level processes.

Critically, self-organization offers a fundamentally different answer to Allport's question. Coherence is not imposed by a trait-like cause (i.e., someone is talkative because a latent Extraversion propensity drives that behavior) but emerges from the ongoing interactions among components of the system. Stability, in this view, is not a fixed property of the person but a dynamic consequence of the system. Regularities arise because internal dynamics constrain behavior over time (i.e., pull the system toward homeostasis), not because they are encoded in a latent structure, which maps directly onto Stagner's (1951) argument for homeostasis as a unifying concept. In this sense, traits are not causes of behavior, but summaries of the stable patterns that self-organizing systems produce. This view preserves a descriptive perspective on traits while also providing an explanatory account of their emergence. Yet, self-organization raises a deeper question: What produces that order?

Self-organization is often misconstrued as implying randomness or spontaneity. Even the simplest dynamical systems demonstrate the opposite: Order emerges from relatively simple,

law-like regulatory forces. A simple linear attractor (Figure 2) illustrates this point. Behavior is not random but systematically drawn toward a stable state by the system's internal dynamics. More complex forms of self-organization build on this same principle, introducing multivariate structures, nonlinearities, multiple attractors (i.e., multistability), and interactions across components.

This alternative conceptualization maps more closely onto everyday psychological dynamics. For example, the lay concept of ambiversion (i.e., individuals who can be highly extraverted but also require withdrawal) is well captured by a system with two attractors separated by a repeller (Danvers et al., 2020; see the second column of Figure 3). Such an individual is unlikely to persist in a moderately extraverted state, as this region is unstable (it *pushes*, rather than pulls). However, when near that point, small contextual shifts can readily push behavior toward either high or low extraversion (e.g., staying out with friends or returning home to be alone). This system would likely produce greater variability in behavior overall, while individuals spend sustained periods in either low or high extraversion before shifting between them.

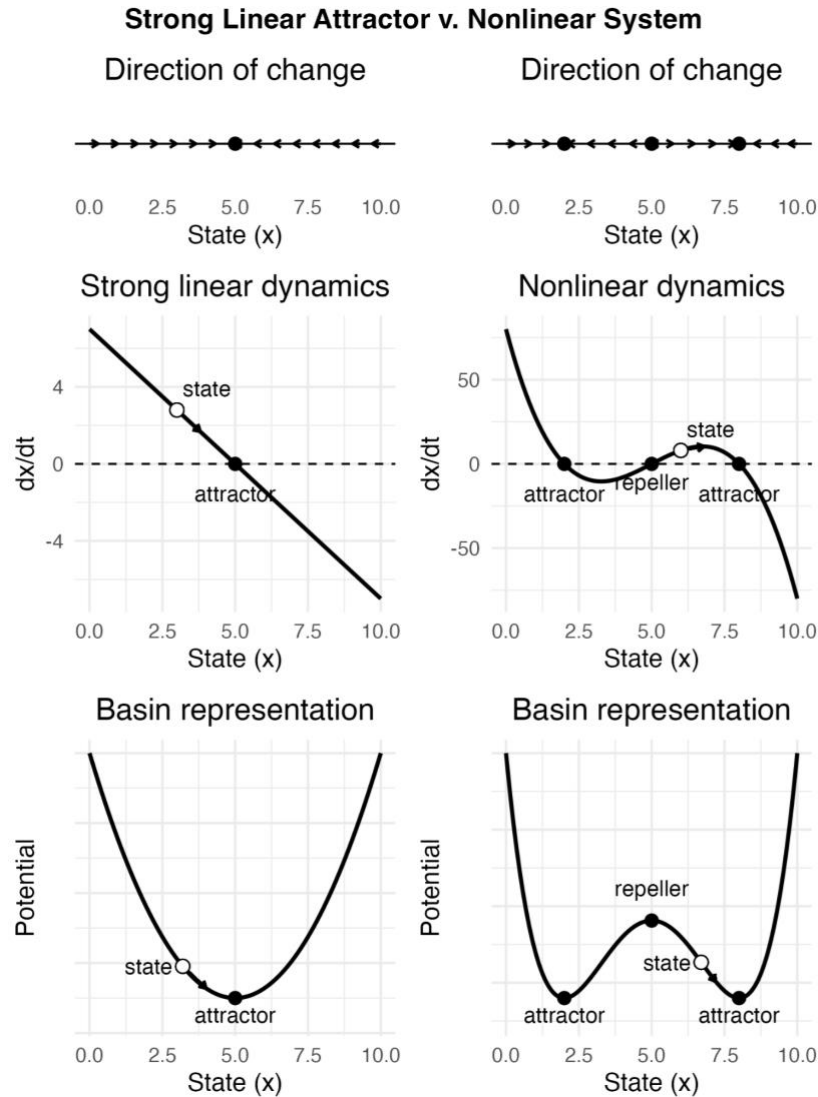


Figure 3. The same underlying systems are visualized in three equivalent ways: direction of change across the state space (top), the state-dependent rate of change (dx/dt ; middle), and the corresponding potential (basin) representation (bottom). The left column depicts a system with a single strong linear attractor, characterized by a constant restoring force toward equilibrium, yielding a single, symmetric basin and rapid convergence from all states. The right column depicts a nonlinear system with multiple fixed points, including two attractors and an intervening repeller. Here, the direction and magnitude of change depend nonlinearly on the current state, producing multiple basins of attraction separated by unstable regions. The white point denotes the current state, the black points denote fixed points (attractors or repeller), and arrows indicate the direction of change implied by the system's dynamics.

Conceptually, multiple attractors are poorly aligned with trait models. Because traits reduce tendencies to a single latent score, they cannot represent systems in which behavior

settles into qualitatively distinct stable states. A trait perspective would collapse the cubic dynamics in Figure 3 into the weak linear attractor shown in Figure 2, assigning equivalent scores to fundamentally different temporal dynamics. Both individuals would be classified as moderately Extraverted, even though the cubic dynamics cannot be accounted for by a latent Extraversion trait as it is typically operationalized. In other words, having a tendency toward moderate levels of Extraversion is distinct from having two distinct tendencies toward high and low levels of Extraversion, and only a dynamic systems perspective can currently differentiate the two. Thus, trait models fail to capture both reactivity and, in nonlinear systems, disposition itself.

This distinction also clarifies an important difference from trait-based approaches that emphasize distributions of behavior (e.g., density distributions; Fleeson & Jayawickreme, 2015). Such approaches characterize where behavior is most likely to occur, often identifying modal regions or variability around a central tendency. In contrast, topological representations not only capture where behavior tends to reside (i.e., attractor basins), but also specify how behavior evolves over time, including whether it is drawn toward, repelled from, or transitions between regions of the state space. Importantly, these dynamics also constrain where the system is likely to have come from, as certain regions of the state space are more or less accessible depending on the system's structure (i.e., transitioning away from a strong attractor requires stronger perturbation than transitioning away from a weaker attractor). Two systems may therefore produce similar distributions of behavior while reflecting fundamentally different dynamics, including differences in stability, reactivity, and the pathways by which states are entered and exited. Distributions summarize *where* behavior has been observed, whereas topologies characterize *how* it unfolds over time.

Critically, the interpretive power of nonlinear attractor models lies not only in the observation of multiple homeostatic states, but in their topology (see Figure 4). While the former lends itself to coarse distinctions (e.g., whether individuals exhibit one or multiple stable states), the latter provides a more precise account of how behavior unfolds over time.

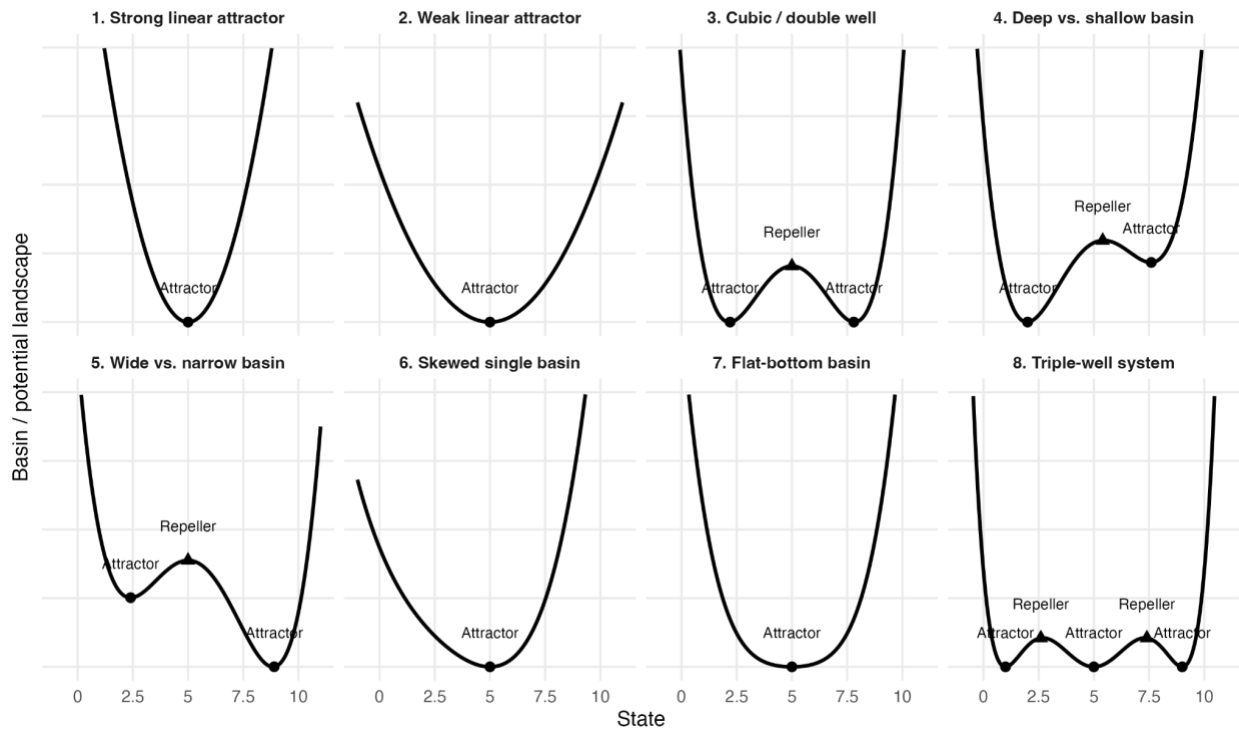


Figure 4. Each panel depicts a potential landscape over state space (x), where valleys correspond to attractors (stable states) and peaks correspond to repellers (unstable states). Panel 1 shows a strong single attractor with steep basin walls, whereas Panel 2 shows a weaker attractor with a shallower basin. Panel 3 illustrates a symmetric double-well system with two attractors separated by a central repeller. Panels 4 and 5 depict asymmetric basin depth, where one attractor is more stable than the other. Panel 6 shows a skewed single basin, and Panel 7 a flat-bottom basin reflecting low restoring forces near the attractor. Panel 8 illustrates a triple-well system with three attractors separated by two repellers. Points mark attractors and repellers.

First, the depth, width, and relative positioning of attractor basins determine whether behavior is stable, variable, or prone to abrupt transitions (Cramer et al., 2016; Van der Maas & Molenaar, 1992). In systems with asymmetrical basins (e.g., Figure 4.4 and 4.5), one state may dominate behavior due to its greater stability, while another remains accessible but fragile

(Lewis, 2005; Wagenmakers et al., 2005). Such systems can produce the appearance of sudden “snapping” or stark behavioral shifts, where small or cumulative perturbations push the system out of a shallow basin into a more stable alternative. For example, in Figure 4.4, a predominantly low-sociability pattern may still include periods of high sociability, albeit less frequently; conversely, in Figure 4.5, a predominantly high-sociability pattern may still include periods of withdrawal. Thus, Figures 4.3, 4.4, 4.5, and 4.8 would all satisfy the verbal description of “ambiversion,” despite reflecting qualitatively distinct dynamics.

Second, rather than linear or nonlinear shifts in trait levels over time, change can be conceptualized as alterations to system topology, including the emergence, disappearance, or reshaping of attractor basins (Beck & Jackson, 2022b; Van der Maas & Molenaar, 1992). This perspective may help explain perplexing patterns in personality development (Warfel et al., 2025), including the inconsistent effects of major life events on trait change, suggesting the need to shift the study of personality trait change to account for person-specific system-level aspects and vulnerabilities (Nissen & Beck, 2024, 2025b). For example, many new parents note feeling like they are forced to increase in Conscientiousness due to increased responsibility, yet there is little replicable empirical evidence of this effect (Bühler et al., 2024; van Scheppingen et al., 2016). One possibility is that new parents’ system typologies shift in ways that are not detectable via trait change, such as shifting from a single, moderately strong attractor to a multistable system with both high and low levels of responsibility. While new parents are forced to be responsible in many key ways, this may come at the expense of reduced responsibility in other moments and domains.

Third, although nonlinear topologies are powerful for describing system dynamics, they remain fundamentally descriptive. They characterize the form of stability in a system (i.e., where

behavior settles, how it shifts, and when it destabilizes) but do not explain why these patterns emerge. In other words, topology specifies how the system unfolds, but not what produces those dynamics (Cramer et al., 2010; Lewis, 2005). Moreover, the examples above are largely univariate, abstracting away from the interactions among cognitive, affective, and behavioral processes that give rise to these patterns. To understand personality as a self-organizing system, we must move beyond univariate topologies to examine complex reciprocal relationships among multiple elements and the regulatory forces that support them.

4.4.1 Multivariate Topologies

One reason these univariate topologies appear nonlinear is that they abstract away from the inherently multivariate nature of the system. As established in the prior section on the whole person, personality is constituted by interacting cognitive, affective, and behavioral components that unfold jointly over time. Representing such a system univariately necessarily collapses these interactions, potentially mischaracterizing the form of the underlying dynamics.

In a systems framework, explanation does not reside in isolated variables or their marginal dynamics, but in the functional relations among interacting components. This distinction reflects a broader shift in how causation is conceptualized. Mechanistic approaches seek separable causes that can be isolated and estimated independently, an assumption that holds when systems can be approximated as linear (Wagner, 1999). In contrast, systems approaches aim to characterize the multivariate functional relations that give rise to system behavior, where components are mutually constitutive and their effects cannot be understood in isolation.

This distinction is illustrated by the difference between what C. J. Lee and colleagues (under review) have termed marginal and joint attractor models. A marginal (univariate) attractor model with an additional process included as a time-varying predictor asks whether the location

or strength of the focal system shifts as a function of that paired variable. In contrast, a joint (e.g., bivariate) attractor model treats both processes as dynamically coupled, allowing the topology itself to emerge from their interaction. As shown by C. J. Lee and colleagues (see Figure 5), marginal attractors can capture whether personality dynamics vary as a function of situational characteristics (e.g., situation perception) but fail to capture how those situational characteristics themselves evolve as part of the system. Joint attractors, by modeling these processes simultaneously, instead characterize the topology of the system as a function of their reciprocal relations, revealing patterns of change in the system that cannot be detected by examining marginal attractors.

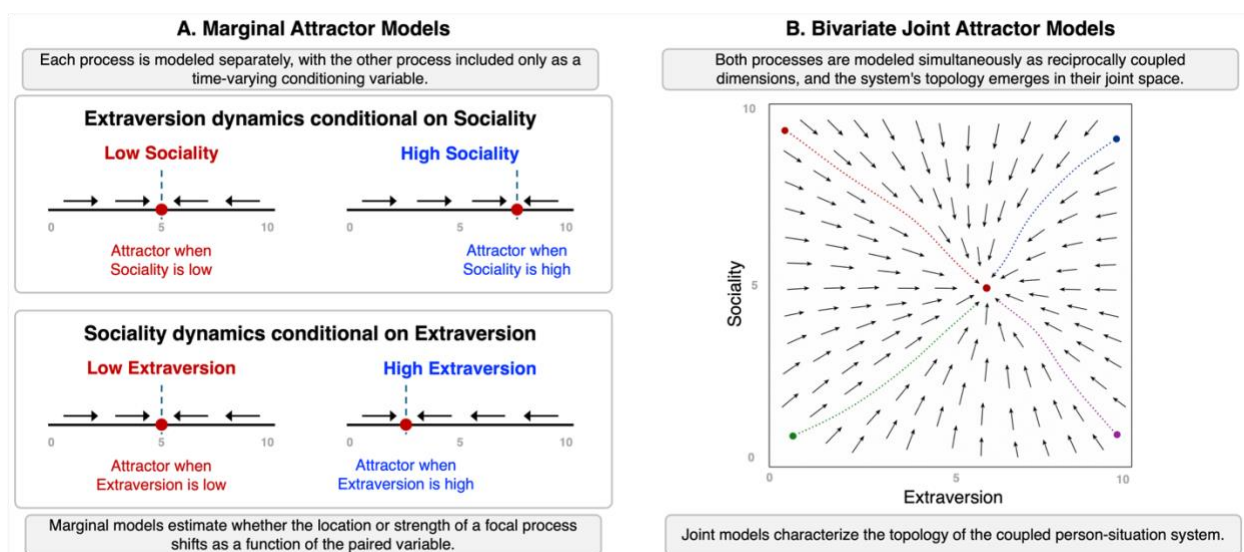


Figure 5. (A) In marginal attractor models, each focal process is modeled separately, with the paired process included only as a time-varying conditioning variable. This permits estimation of whether the location of a univariate attractor shifts as a function of the other process (e.g., whether Extraversion homeostasis differs under low versus high Sociality, or vice versa, but does not represent the conditioning process as dynamically evolving within the same system. (B) In joint attractor models, both personality states and situational characteristics are modeled simultaneously as reciprocally coupled dimensions, yielding an emergent vector field in their shared state space. As a result, the topology of the coupled person-situation system, including trajectories of change and homeostatic structure, arises from the reciprocal relations between processes rather than from isolated conditional slices.

In sum, univariate models provide useful summaries of whether a system exhibits attractor-like properties, and nonlinear extensions capture important features such as multiple homeostatic states. However, these approaches remain limited in their explanatory scope. The personality system is fundamentally multivariate, and failing to model it as such constrains our ability to understand the processes that generate self-organization and emergence. Critically, considering a multivariate system begins a shift from describing topological features to identifying the endogenous and exogenous regulatory forces that govern interactions among components. It is these forces that generate, maintain, and transform the topology of the system.

4.5 Regulatory Forces in Dynamic Systems

“I have learned that while the major foci of organization in a life may be few in number, the network of organization, which includes both minor and contradictory tendencies, is elusively complex” (Allport, 1968, p. 63).

To meet the criteria of an adequate model of personality, a framework must not only describe patterns of behavior but explain why those patterns emerge and persist. In dynamic systems terms, this requires identifying the regulatory forces that generate and constrain system dynamics. Moving beyond topology (e.g., attractor basins) to explanation therefore requires specifying the forces that produce those dynamics in the first place.

A useful distinction is between endogenous and exogenous regulatory forces. Endogenous forces arise within the system and include processes such as goals, affective tendencies, learned associations, and feedback among cognitive, emotional, and behavioral components. Exogenous forces originate outside the individual and include situational demands, social interactions, cultural norms, and structural conditions. In practice, these forces are deeply intertwined: system dynamics reflect the ongoing coupling between processes inside the person and constraints outside the person. For example, a social demand matters not only because it

exists in the environment, but because it is perceived, interpreted, and acted upon by the individual; likewise, internal goals and affective experiences unfold within concrete situations that afford some actions and constrain others.

A hallmark feature of complex systems is that highly structured dynamics can emerge from relatively simple interacting forces (e.g., Nowak et al., 2005). For example, coordinated bird flight can arise from a small set of local rules (e.g., separation, alignment, and cohesion), yet produces coherent, emergent patterns at the group (or flock) level. Analogously, patterns of personality expression may arise from the interaction of a limited set of regulatory processes operating across levels, even as the resulting dynamics appear complex and individualized.

Importantly, our goal here is not to provide a comprehensive account of each conceivable class of regulatory force, which could span a volume on its own. Rather, we aim to illustrate how different types of constraints (see Table 4) shape system dynamics and to highlight the need for integrative models that bring these forces together. Each of these domains has been studied extensively; the challenge moving forward is not their identification, but their integration within a unified dynamical framework.

Table 4*Example Regulatory Forces of Psychological Systems and Their Possible Constraints*

Category	Examples	How it constrains the system	Timescale / stability
Biological	Genetics, hormones, blood-based biomarkers, chronic health conditions	Sets boundary conditions on possible states; influences reactivity and recovery	Relatively stable to moderately changing; from momentary physiological shifts to lifespan-level constraints (Hawks et al., 2024; Schwaba et al., 2025)
Goals	Personal goals, motivations	Directs trajectories, shapes how processes unfold and are maintained	Span nested timescales, from momentary intentions to medium-term personal strivings to long-term life goals; stability generally increases with goal abstraction, identity centrality, and commitment (Carver & Scheier, 1982; Emmons, 1997)
Values / norms	Moral beliefs, internalized standards, expectations	Shapes evaluation of states; reinforces or suppresses recurring patterns	Typically operate over longer timescales, from enduring internalized standards to culturally shared norms; relatively stable, but can shift or dramatically change through socialization, identity change, or sustained contextual pressure (Bardi & Goodwin, 2011)
Situational	Immediate contexts, affordances	Perturbs system; triggers transitions between states	Typically short-term and rapidly changing; momentary psychological contexts and affordances can shift quickly with changes in environment or its perception (Rauthmann & Sherman, 2020)
Interpersonal	Relationships, social feedback, reputation	Creates recurring patterns or feedback loops; stabilizes or destabilizes patterns	Ranges from momentary interactions to enduring relational structures (A. G. C. Wright et al., 2023)
Cultural	Norms, institutions, goals	Defines broad constraints on system organization (acceptable goals, behaviors, and self-concepts)	Long-term and highly stable at the individual level (Lehman et al., 2004)
Socio-structural	Resource access, systemic affordances and barriers, policy, norms	Identifies the degree and kinds of broad constraints on system organization for individuals of varying social status	Timescale varies between short (e.g., situational and interpersonal) to long-term (e.g., resource access); generally stable but is dependent on shifts in sociopolitical regimes (Watts et al., 2003)
Multi-level	Interactions across above	Constraints interact nonlinearly across levels (e.g., top-down constraints)	Reflects timescales and stability across levels (Bronfenbrenner, 1977)
Habits	Repeated behavioral sequences, daily routines, scripts	Narrows the range of likely responses by increasing automaticity and recurrence	Medium to long-term once formed; often stable unless disrupted (G. W. Allport, 1937; W. Wood & R�nger, 2016)

Identity	Identity commitments, role identities, personal narrative, self-perception motives	Organizes coherence across time and (partially) determines which states feel congruent or incongruent	Like values, typically operates over longer timescales, though revisable (McAdams et al., 2021; McAdams & McLean, 2013)
Affective regulation (of the system)	Responsiveness to feedback, reward/threat sensitivity	Alters elevation, dampening, and recovery dynamics; changes persistence of states	Short-term operation with medium term individual differences in stability (Block, 2002; Kuppens & Verduyn, 2015)

Across domains, regulatory forces operate by constraining and reshaping the system's topology, altering which states are accessible, how stable they are, and how the system transitions between them. In this sense, regulatory forces are not additional variables to be included in a model but the mechanisms that give the system its structure in the first place. Different constellations of regulatory forces can therefore produce distinct topologies, with implications for multifinality, equifinality, multicausality, and nonergodicity in personality processes.

For example, biological processes may set boundary conditions on possible states and influence reactivity; goals may direct trajectories and sustain patterns over time; situational inputs may perturb the system and trigger transitions; and interpersonal and cultural contexts may stabilize or destabilize recurring patterns through feedback and reinforcement. Critically, these forces do not operate in isolation, but interact across levels, jointly shaping the dynamics of the system.

Taken together, this perspective shifts the explanatory task from describing topological features to identifying the regulatory forces that generate, maintain, and transform them. In doing so, it provides a foundation for understanding how self-organization arises in personality systems and sets the stage for a more detailed account of the mechanisms through which these forces operate.

This perspective also provides a natural account of why broad trait-like structures, such as the Big Five, are so robust at the population level. Rather than reflecting fixed underlying causes that operate identically within each person, these dimensions may arise as higher-order regularities across persons whose systems are shaped by partially shared biological, social, cultural, and environmental constraints. For example, recurrent constraints related to social

connection, goal pursuit, and environmental demands may give rise to common patterns of sociability, emotional reactivity, and behavioral regulation across individuals in a given culture.

This interpretation differs from Cybernetic Big Five Theory (CB5T), which treats the Big Five as core organizing dimensions grounded in biological and cybernetic systems (DeYoung, 2015). In contrast, a regulatory forces framework treats these dimensions as emergent outcomes of interacting constraints. From this perspective, shared biological and motivational systems may make certain structures likely, but they do not fix them (Lewis, 2005; Vallacher et al., 2002). Instead, the structure that emerges depends on how these forces interact with cultural, situational, interpersonal, and cognitive constraints (Magnusson, 2001; Read et al., 2010; Vallacher et al., 2002). As a result, variation in these forces across contexts may produce meaningful differences in the structure of personality itself, rather than variation within a fixed set of dimensions (Beck & Jackson, 2020a; Beltz et al., 2016).

A related implication concerns the role of characteristic adaptations. Within CB5T, much of the variability associated with goals, beliefs, and situational processes is treated as secondary to the core trait dimensions, often conceptualized as contextualized expressions of an underlying system (DeYoung, 2015; McCrae & Costa, 2008). From a regulatory forces perspective, however, these processes are not peripheral, but constitutive. Variation in goals, beliefs, social contexts, and cultural constraints reflects meaningful differences in the forces shaping the system, not noise around a stable structure (D. Wood et al., 2015). This distinction becomes especially important in multi-level systems, where interactions among regulatory forces across levels can produce qualitatively different system dynamics and, consequently, different emergent structures (Magnusson, 2001; Read et al., 2010; Vallacher et al., 2002).

4.5.1 *Multi-level system dynamics*

Regulatory forces are particularly complex because psychological systems are inherently multi-level. Momentary affect aggregates into emotion, mood, and broader dispositions, which in turn scale to group- and cultural-level patterns. As open systems, these levels mutually influence one another. An example of a three-level multi-level system is depicted in Figure 6.

In self-organizing systems, higher-level dynamics act as constraints on lower-level processes, shaping which states are accessible, how stable they are, and how the system transitions between them (Ashby, 1947; Erofeeva et al., 2024; Juarrero, 1999, 2023). Although these processes are often studied using multilevel models, which represent one of the most advanced tools for capturing cross-level structure in psychological data, such approaches primarily capture linear moderation of lower-level processes that are often modeled as linear themselves (Raudenbush, 2002; Snijders & Bosker, 2011). A dynamical perspective instead suggests that higher-level processes reorganize the system's topology, not merely its current state.

This distinction is critical. In multilevel models, higher-level variables are typically treated as contextual modifiers that shift the strength or direction of lower-level associations (Raudenbush, 2002). In contrast, a systems perspective treats higher-level processes as constraints that reshape the organization of the system itself, altering which states are possible, their stability, and how the system evolves over time (Juarrero, 1999; Kelso, 1995; Read et al., 2010).

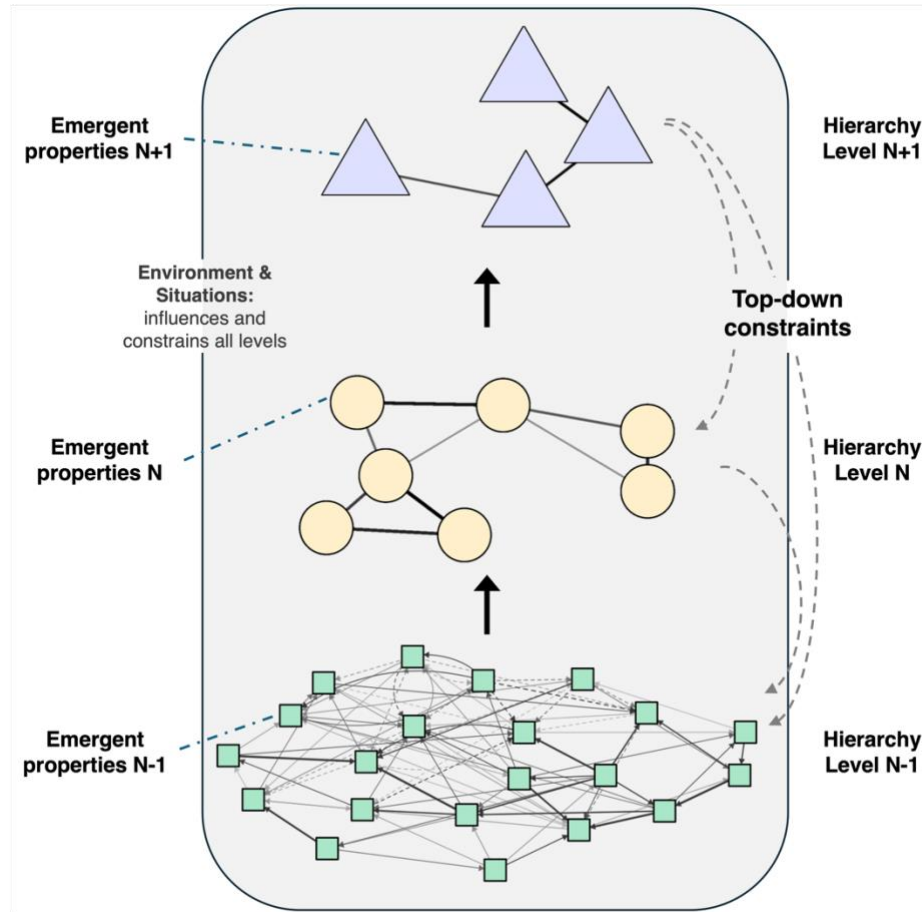


Figure 6. Conceptual illustration of multi-level organization in a dynamic system across three hierarchical levels (N-1, N, N+1). At the lowest level (N-1), elements are linked through directed, dynamic relations. Higher levels (N, N+1) represent increasingly abstract patterns of organization that emerge from these interactions without assuming one-to-one mappings between levels. Upward arrows denote the emergence of higher-level structure, whereas dashed curved arrows denote top-down constraints that shape dynamics across levels. The enclosing boundary represents the broader environment, which constrains and interacts with system processes at all levels. This schematic provides a general framework that can be instantiated differently across individuals and extended to higher-order structures.

This perspective has important implications for understanding situations and culture.

Situations are not simply external inputs, but perturbations that interact with ongoing system dynamics, often becoming embedded within feedback loops that stabilize or destabilize patterns over time (Mischel & Shoda, 1995; Vallacher et al., 2002). Similarly, cultural systems operate as higher-level regulatory forces that constrain the organization of lower-level processes, shaping

which states are reinforced and how individuals interpret and respond to their environments (Markus & Kitayama, 2010).

Taken together, a multi-level perspective highlights that regulatory forces do not operate independently but interact across levels to generate system dynamics. Understanding personality therefore requires modeling how constraints at different levels jointly shape the structure and evolution of the system.

4.5.2 *Regulatory forces as unifying concepts*

Seventy-five years ago, Ross Stagner (1951) proposed homeostasis as a unifying concept in personality. However, what shapes those homeostatic points is an even more fundamental question, suggesting that regulatory forces may serve as a more basic organizing principle.

Regulatory forces are not simply omitted or moderating variables to be added into existing models. Rather, they are the mechanisms that give psychological systems, including their homeostatic points, their structure in the first place (Juarrero, 1999; Kelso, 1995; see Figure 6). By constraining which states are accessible, how stable they are, and how transitions occur, these forces generate the topology of the system and give rise to stable yet flexible patterns of thought, feeling, and behavior (Kelso, 1995; Lewis, 2005).

Personality, in turn, can be understood as an emergent hierarchy shaped by interacting regulatory forces that operate both internal and external to the person and across levels of analysis. This emergent hierarchy accounts for phenomena such as multifinality, equifinality, multicausality, and nonergodicity, which have long posed challenges for traditional approaches (Cicchetti & Rogosch, 1996; Molenaar, 2004). It also clarifies that inquiries into the role of domains from genetics to situations in shaping personality were not misplaced but erred instead

in treating these domains as separable mechanistic causes of behavior rather than as interacting regulatory forces that generate and constrain system dynamics.

Broad trait-like dimensions such as the Big Five may therefore be understood as recurrent higher-order regularities produced by shared constellations of regulatory forces, rather than as independent mechanistic components of personality (Fleeson & Jayawickreme, 2025; Revelle & Condon, 2015). From this perspective, sociability, emotional reactivity, and behavioral regulation are not separable causal entities to be parsed apart, but mutually constraining processes that co-evolve through their functional relationships (Read et al., 2010; Vallacher et al., 2002). Because humans share common pressures related to social connection, goal pursuit, and environmental adaptation, these interacting forces may often yield recognizable patterns of sociability (Extraversion/Agreeableness), emotional reactivity (Neuroticism), and behavioral regulation (Conscientiousness) across individuals. Importantly, however, such regularities do not imply a fixed or invariant underlying structure (Juarrero, 1999). Instead, they reflect the provisional organization that emerges when similar regulatory constraints operate within comparable developmental and cultural contexts, which may explain why some population-level traits, like Openness, fail to replicate across cultures (Schwaba & Thalmayer, 2026). Dynamic systems approaches are uniquely positioned to capture this balance of recurrence and contingency, as they explicitly model how interacting constraints generate system organization, stability, and change over time. In this way, they provide a framework not only for describing personality, but for explaining why broad personality regularities emerge at all.

4.6 Feedback Loops and Changes to a System

“Personality is less a finished product than a transitive process. While it has some stable features, it is at the same time continually undergoing change” (Allport, 1955, p. 19).

A dynamic systems perspective reframes personality change in two fundamental ways: What it means for personality to change, and the mechanisms through which such change occurs.

First, personality change is not limited to shifts in mean levels or rates. Although such changes can be represented within system topologies (e.g., as changes in the locations of basins), they are more appropriately understood as consequences of deeper alterations to the system itself. From a systems perspective, change reflects transformations in how the system unfolds over time, including shifts in attractor strength, the emergence or disappearance of attractors, and changes in the number and stability of homeostatic points (i.e., a change in the system's topology; Beck & Jackson, 2022b; Van der Maas & Molenaar, 1992). Rather than assuming monotonic, linear change, this perspective allows for discontinuities and qualitative reorganization. For example, social investment theory posits that personality change follows from the adoption of new social roles and identities (Roberts et al., 2005). Becoming a parent, for example, is unlikely to produce a simple linear increase in "conscientious" behaviors; instead, it may give rise to a new attractor or alter the existing topology in ways that support adaptation to new environmental demands for raising a child. In this sense, personality change is best understood as a reorganization of system structure rather than a shift along a fixed dimension. Importantly, the likelihood and magnitude of such reorganization depend on the stability of the existing system: more stable topologies (e.g., deeper attractor basins) are more resistant to perturbation, whereas less stable configurations are more susceptible to qualitative change (Kelso, 1995).

Second, the mechanisms of personality change are located in endogenous and exogenous regulatory forces that shape how the system evolves over time. These include goals, reinforcement and learning processes, situational inputs, and broader cultural influences (Jackson

& Wright, 2024). Critically, these forces operate through feedback loops, such that past states constrain current dynamics and repeated patterns of activation progressively stabilize or destabilize system configurations (Wrzus & Roberts, 2017). Learning, in this context, reflects the gradual tuning of system parameters and topological features through experience, such that similar inputs may produce different outputs over time as the system reorganizes (Lewis, 2005; Thelen & Smith, 1994).

Importantly, these two features, topological reorganization and feedback-driven dynamics, are inseparable. The impact of any perturbation depends on the current configuration of the system, and feedback processes determine whether that perturbation is amplified, dampened, or integrated into the system's structure (Juarrero, 1999; Kelso, 1995). For example, two emerging adults who leave their parental homes to attend college may encounter similar cultural shifts, but the extent to which these perturbations lead to reorganization will depend on the interaction between these cross-level cultural forces and the individual's existing system topology. Imagine that these two individuals have equivalent average tendencies to invest effort in school-related tasks, translating to similar scores on Big Five Conscientiousness but different underlying structures: Person A is characterized by a single, relatively shallow attractor basin, and Person B by two deeper basins corresponding to distinct behavioral modes (e.g., highly organized vs. disengaged states). Person A may exhibit more flexible but less stable dynamics, readily shifting in response to environmental demands, whereas Person B may become "trapped" in one mode for extended periods. Each person may respond similarly to situational and cultural forces that reinforce relatively high sustained academic effort (e.g., adequate structure, resources, clear expectations), resulting in a deepening of this attractor basin and similar levels of downstream success for both individuals. However, greater exposure to forces that detract from

academic effort may have drastically different consequences – whereas Person A might bounce back but ultimately become more productive across time, Person B might get stuck in a prolonged state of disengagement and thus experience a deepening of that attractor basin instead, leading to drastic differences in life outcomes relative to Person A. Thus, similar environmental inputs can produce qualitatively different developmental trajectories between Person A and B, not because both differ in mean levels, but because they differ in the structure and stability of the systems through which those inputs are processed (Beck & Jackson, 2022b; Cicchetti & Rogosch, 1996; Molenaar, 2004). This perspective aligns with prior work highlighting the role of selection effects in shaping exposure to life events (e.g., Beck & Jackson, 2022a), but extends it by emphasizing that change processes themselves are nonlinear and person-specific, rather than primarily linear shifts in mean levels or covariation among indicators.

A substantial body of empirical work examining life events as mechanisms of personality trait change has produced notably inconsistent findings (e.g., Bühler et al., 2024). In many cases, apparent socialization effects diminish or disappear once selection effects (i.e., more conscientious people are more likely to be parents) are accounted for (e.g., Bollich-Ziegler et al., 2021; Hill et al., 2021; van Scheppingen et al., 2016), and even when change is observed, it is often modest and difficult to replicate across samples. From a dynamic systems perspective, however, such inconsistency is expected rather than problematic. Because change is contingent on person-specific system topologies, unfolds nonlinearly, and is shaped by ongoing feedback processes (e.g., Cicchetti & Rogosch, 1996; Molenaar, 2004), similar environmental inputs need not produce similar outcomes, reflecting multifinality. What appears as empirical inconsistency under traditional paradigms may therefore reflect lawful variability in system reorganization (Juarrero, 1999).

Accordingly, a dynamic systems framework suggests a new agenda for studying both lifespan and volitional personality change that integrates processes across levels, from cultural and cohort influences (e.g., Atherton et al., 2025) to the micro-dynamics of within-person system organization. Critically, this perspective emphasizes that such changes emerge bottom-up via self-organization within a multi-level system.

4.7 Linking Levels in the Personality System

“Whatever individuality is, it is not the residual ragbag left over after general dimensions have been exhausted. The organization of Bill’s life is first, last, and all the time, the primary fact of his human nature” (Allport, 1968, p. 88).

Allport’s insight anticipates a central challenge in contemporary personality science: How to formally account for the organization of the person as a multi-level system. Any adequate model of personality must therefore explain how processes at different levels of aggregation and analysis relate to one another. Psychological phenomena span multiple levels, from biological and physiological processes, to preconscious and cognitive-affective dynamics, to momentary states, to broader traits and narrative identity, and ultimately to cultural systems (Magnusson, 2001). Historically, these levels have often been studied in isolation or linked through reductionist or top-down frameworks that treat higher-level constructs as primary causes (Magnusson, 2001), even as other approaches have emphasized bidirectional influences across levels (e.g., Bronfenbrenner, 1977).

A dynamical systems perspective offers an alternative: Levels are linked through processes of self-organization that give rise to an emergent hierarchy (Juarrero, 1999; Revelle & Condon, 2015). Lower-level processes, which are often heterogeneous, interacting, and time-varying, *produce* organized, stable patterns over time, which are observed as emergent higher-level constructs such as traits (Kelso, 1995; Read et al., 2010). Critically, these higher-level

patterns do not exist independently of the processes that produce them. Rather, they summarize and constrain ongoing dynamics, shaping the range of states the system can occupy. This proposal is not merely conceptual. Existing techniques for modeling psychological systems (e.g., Chen et al., 2025; Lewis, 2000; Peck, 2009) and other complex systems (e.g., Erofeeva et al., 2024) demonstrate the feasibility of multi-level dynamic approaches.

From this perspective, personality is not a collection of separable levels that can be ranked in terms of incremental validity, nor does any single level constitute the “true” level of personality. Instead, personality is a multi-level system in which structure emerges from interactions across time and scale (Baumert et al., 2017; Cramer et al., 2012). Consequently, no unit of personality – from momentary processes to states, traits, or narrative identity – exists as a fixed entity independent of the dynamics that produce it. Rather, these constructs are emergent properties of ongoing, multi-level processes. Understanding personality therefore requires modeling how dynamics unfold within and across levels, rather than privileging any single level of analysis. Figure 7 summarizes candidate levels in this emergent personality hierarchy and illustrates the recursive pathways through which they constrain and stabilize one another.

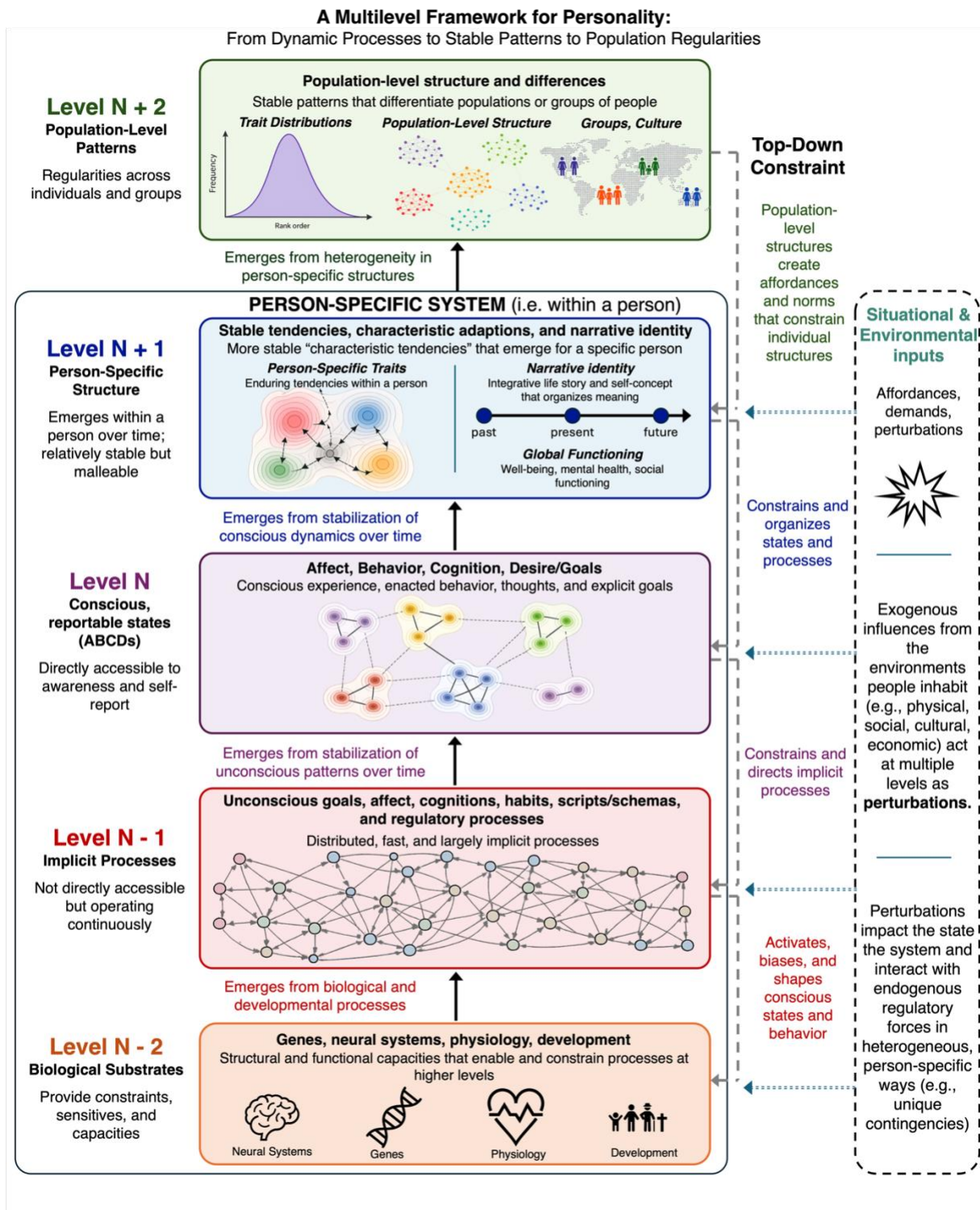


Figure 7. Personality is conceptualized as a hierarchy of interdependent levels spanning biological substrates (Level N-2), implicit processes (Level N-1), conscious states and behaviors (Level N), person-specific structures (Level N+1), and population-level regularities (Level N+2). Higher-level organization emerges from the dynamic interactions of lower-level processes, with resulting structures stabilizing and constraining those underlying dynamics over

time. Within individuals, repeated interactions among affect, behavior, cognition, and desires/goals (ABCDs) give rise to stable, person-specific structures (e.g., characteristic tendencies and narrative identity) through processes of self-organization and stabilization over time. Across individuals, heterogeneity in these person-specific systems produces emergent population-level patterns (e.g., trait distributions, network structures, and cultural differences), which in turn create social affordances and norms that recursively shape individual systems. This framework emphasizes that personality is not reducible to any single level, but instead reflects a dynamically coupled, multi-level system in which structure and process co-emerge across timescales.

However, it is important to again distinguish between modeling multiple levels of nested data structures and modeling a multi-level system. Multilevel models (MLM) are well-suited for capturing statistical relationships across levels (Raudenbush, 2002), but they do not, by themselves, specify how interactions among lower-level processes give rise to higher-level structure. In particular, cross-level effects in MLM are typically represented as linear associations or moderators, rather than as changes in system topology. As a result, modeling levels is not equivalent to modeling the emergent, dynamically organized system that constitutes personality.

While modeling multi-level systems across levels and timescales is increasingly possible, modeling the emergence of higher-level structure from lower-level processes remains an open challenge (Hoel et al., 2013; Jansma & Hoel, 2025). Existing approaches can capture dynamics within levels and, in some cases, characterize cross-level associations, but they do not yet “engineer” emergence in the sense of deriving higher-level organization directly from first principles. Instead, current models are best understood as being consistent with emergent processes rather than fully specifying them (Van Orden et al., 2003). This limitation is not unique to personality science but reflects a broader challenge in the study of complex systems (Bedau, 1997; Mitchell, 2009). Advancing our understanding of personality as a multi-level, self-organizing system will therefore require continued development of models that explicitly link

dynamics across levels, rather than treating higher-level structure as fixed or given. Critically, emergence in personality systems is fundamentally bottom-up, underscoring the importance of capturing the dynamics of individual lives as they unfold within their contexts before imposing higher-level structure or consensual dimensions (Nissen & Beck, 2025b). We build on this perspective in the final section by outlining a framework for understanding personality through bottom-up integration.

4.8 Summary: Assessing the Adequacy of Person-Specific Nonlinear Dynamic Systems

In this section, we outlined the fundamental features of an adequate model of personality that accounts for characteristic patterns of thinking, feeling, and behaving; explains both stability and change; incorporates context; captures the coherence of the whole person; and accommodates core properties of psychological systems such as non-ergodicity and nonlinearity. We argued that a person-specific nonlinear dynamic systems framework meets these criteria where other models fall short, providing an integrative account that spans levels of analysis, from persons to populations and from momentary processes to developmental timescales.

A dynamic systems perspective advances these criteria in several key ways. It reconceptualizes characteristic patterns as emergent properties of system dynamics, accounts for both short-term fluctuations and long-term stability and change through system topologies, and explicitly incorporates contextual influences via open systems and regulatory forces. It further reframes coherence as an emergent product of self-organization and offers a principled account of the whole person by linking levels of analysis through emergence and the constraints that give rise to it.

At the same time, this framework satisfies key epistemological and formal requirements, including non-ergodicity, multifinality, equifinality, multicausality, and nonlinearity, that have

long posed challenges for traditional approaches. In doing so, it both accommodates the strengths of existing paradigms and addresses their core limitations, positioning it as a coherent foundation for cumulative personality science.

An important question, however, is what would count as evidence against such a framework. Although we do not propose a single formal theory, a person-specific dynamic systems perspective entails a set of empirical commitments that are, in principle, falsifiable. First, it assumes that psychological processes are generally non-ergodic, such that within-person structures do not reduce to population-level regularities (Molenaar, 2004). Evidence that most psychological processes are approximately ergodic (i.e., population-level models accurately recover within-person dynamics) would challenge this assumption. Second, it implies that person-specific models should provide incremental value over population-level approaches in describing and predicting behavior (Beck & Jackson, 2022c; Soyster et al., 2021). If general models perform equally well across individuals, the need for person-specific inference would be undermined. Third, the framework assumes that psychological phenomena arise from multivariate, interacting processes rather than a small set of independent latent causes (Beck & Jackson, 2020a; Cramer et al., 2012; Mischel & Shoda, 1995). Evidence that invariant latent structures more fully account for behavior across persons and contexts would call this assumption into question. Finally, the framework assumes that psychological dynamics are often nonlinear and may involve multiple stable states or topological reorganization (Cramer et al., 2016; Lewis, 2005; Van der Maas & Molenaar, 1992). If psychological processes are adequately described by linear, single-equilibrium models, the added complexity of a systems perspective would not be warranted. Taken together, these considerations clarify that, while broad in scope, a

dynamic systems framework makes concrete empirical claims that can, in principle, be disconfirmed.

In practice, its primary limitation lies in implementation. Although the formal and methodological tools required to operationalize this perspective are increasingly well developed, they remain distributed across literatures and are not yet widely integrated into standard research practice. A related challenge lies in data: Adequately capturing the dynamics of personality systems requires dense, temporally resolved measurements at the appropriate level of analysis, which are often difficult to obtain in practice. Importantly, these limitations reflect not shortcomings of the framework itself but constraints in the current state of methodological tools and data collection. Accordingly, we do not attempt to fully resolve these challenges here. Instead, we turn in the remainder of the chapter to the conceptual and practical implications of adopting a dynamic systems paradigm for personality psychology – and for psychology more broadly – while directing readers to existing work for detailed methodological guidance.

Taken together, this framework provides a conceptually and empirically accountable foundation for rethinking personality as a person-specific dynamic system. Critically, it also reframes how cumulative knowledge in personality science is achieved: Not through aggregation across individuals assumed to share a common structure, but through the emergence of population-level regularities from systematic comparisons across person-specific systems. In what follows, we consider the conceptual and practical implications of adopting this perspective for personality psychology and for psychology more broadly.

5 Implications of a Person-Specific Systems Perspective

“In contrast to empirical research, our intuitions operate on idiographic rather than nomothetic assumptions” (Bem & Allen, 1978, p. 510).

Thus far, we charted the historical progression of personality theory, noting how the trait approach's ubiquity emerged from pragmatic constraints, available methodological approaches, and historical events. We have argued that population-level approaches are epistemologically insufficient as models of personality, and that a nonlinear, person-specific dynamic systems perspective provides a more holistic framework, one that captures personality as a process of self-organization and emergence across multiple levels unfolding within individuals over time.

In this section, we turn to the implications of this perspective. Rather than continuing to treat the individual as a noisy instantiation of population-level structure, which Allport (1968) described as “hastening to the general,” we argue that personality science must invert its explanatory priority. The person, not the population, must be the primary unit of analysis, requiring instead that we “hasten to the internal pattern” (Beck & Jackson, 2020b; Beltz et al., 2016; Fisher et al., 2018; Lamiell, 1981; Molenaar, 2004; A. G. C. Wright & Woods, 2020). From this vantage point, generalizable knowledge is not imposed top-down through aggregation but emerges bottom-up through the systematic integration of person-specific processes.

Importantly, these implications extend beyond personality psychology to all branches of psychology, including social, cultural, and developmental, and apply across quantitative and qualitative traditions (Borsboom et al., 2003; Cronbach & Meehl, 1955; Molenaar, 2004; Nesselroade et al., 2007; Saqr et al., 2024). This is because the challenge is not specific to any one area or method but arises whenever the goal is to characterize individuals using structures that generalize across persons. To the extent that approaches, whether quantitative or qualitative, rely on identifying recurring patterns, shared themes, or common dimensions, they necessarily privilege what is typical and attenuate what is rare or person-specific (Lamiell, 1981, 1998; Salvatore & Valsiner, 2010; Uher, 2025). Fundamentally, understanding how psychological

processes unfold therefore requires starting with the person and proceeding to the population via bottom-up integration.

This perspective should not be taken to imply that psychologists must now define fully specified, highly multivariate systems across all possible levels in order to make any inference. Such an expectation would far exceed the limits of current tools, data, and theoretical development. Rather, our goal is to encourage a shift in how psychological systems are conceptualized, measured, and analyzed – toward approaches that are more explicitly aligned with the multi-level, dynamic, and person-specific nature of the phenomena they aim to capture. In this sense, the aim is not exhaustive specification, but commensurability: Ensuring that the level of analysis and the inferences drawn are appropriately matched to both our theories and the epistemological constraints of the science (Borsboom et al., 2004).

Accordingly, we begin by considering the elements of the system, arguing that nonergodicity and a systems perspective render isomorphic measurement strategies insufficient (Nissen & Beck, 2025b). We next outline approaches capable of flexibly identifying and integrating personalized indicators across individuals and levels of analysis, before turning to the structure of the multi-level personality system, including both the number of levels and the nature of their interconnections.

5.1 What are the elements of the system?

“The mere cataloguing of ingredients defines personality no better than the alphabet defines lyric poetry” (Allport, 1937, p. 44).

Up to this point, we have largely distinguished elements in terms of levels of aggregation (e.g., processes, states, traits), timescales (e.g., momentary, daily, lifespan), or domains (e.g., affect, behavior, cognition, desire, and environment). From a systems perspective, however,

there is no single privileged answer to what the elements are; elements are defined relative to the level at which the target system is specified.

For example, Figure 7 illustrates a central implication of a person-specific systems perspective: Personality is not localized to a single privileged level of analysis but unfolds across nested levels ranging from biological substrates and implicit regulatory processes to conscious states, person-level structures, and population regularities. Thus, asking what the elements of the system are does not imply a single universal answer independent of the system under study. Rather, elements are defined relative to the level at which the focal system is specified.

Importantly, this does not imply that some levels are more “truly” personality than others, nor that personality science should continue adjudicating which level provides the greatest incremental validity. Biological sensitivities, implicit regulatory processes, momentary affective, behavioral, cognitive, and motivational states, enduring person-specific structures, and broader population regularities are all legitimate components of personality when personality is understood as a multi-level dynamic system. The relevant question is therefore not which level is primary in an absolute sense, but which level is serving as the explanatory target in a given investigation and what kinds of inferences that level allows.

Once this target level is specified, the elemental units of the system are the interacting components operating within that level, whereas adjacent levels function as emergent constraints or enabling conditions. For example, if the target system concerns daily emotional dynamics, then momentary affects, goals, cognitions, behaviors, and situational affordances may serve as target processes, whereas neural activity functions as a lower-level substrate and person- or population-level trait summaries function as higher-level emergent regularities. Neural processes are not irrelevant, nor are trait summaries invalid; rather, neither constitutes the elemental

organization of the focal system unless the system is explicitly redefined to incorporate those cross-level linkages.

This flexibility is a strength, but it comes with an important inferential constraint: The level at which a system is defined determines not only the scope of valid explanation but which elements and levels are available for inference at all. If a level is not explicitly modeled and linked within the target system, it cannot simply be inferred into explanation. We cannot infer momentary within-person dynamics from daily aggregates, person-level organization from group means, or subgroup processes from cultural regularities without demonstrating the structure that connects those levels exhibit ergodicity. Absent such specification, these inferences do not reflect the organization of the system but instead import a structure with presumed universal elements and dynamics from outside it.

Critically, such linkage is not achieved simply by linearly aggregating lower-order observations or by treating between-person differences in average within-person effects as direct representations of higher-order structure. These approaches preserve the statistical form imposed by the analyst, but may fail to capture the nonlinear, heterogeneous, and emergent organization through which higher-level personality structures arise. Cross-level inference therefore requires explicit modeling of how regularities stabilize, transform, or constrain processes across adjacent levels, rather than assuming that summary parameters are isomorphic with the system's organization.

In experimental and aggregate research traditions, the issue of levels of inference is particularly salient because they are often interpreted as revealing the processes by which individuals operate. Formally, however, such designs typically establish only that a manipulation changes an outcome on average across persons. This is not equivalent to demonstrating that the

same process operated within individuals, nor that it operated similarly across them (Lamiell, 2003). As Grice and colleagues (2020) similarly argued, aggregate effects may obscure substantial heterogeneity in the proportion and form of person-level responses. In this sense, as Allport once wrote, “The method required it, but the life did not” (G. Allport, 1968, p. 62).

Accordingly, *if higher-level constructs are to support valid inference about lower-level processes, they must be built from them*. Explanation must proceed bottom-up: Beginning with the level at which the relevant processes unfold and integrating upward to construct higher-order representations that retain principled links to their constituent dynamics. Assuming that higher-level constructs translate automatically into lower-level elemental processes risks masking the very organization that defines personality within persons (Long, Lewis, et al., 2026; Nissen & Beck, 2025b). If we seek to make inferences about individuals from population-level patterns, we must begin with the person-specific system and specify the levels that connect outward from it; otherwise, the individual remains unavailable to explanation.

5.2 How many levels are in the system?

“The psychologist who [studies one function at a time] should recognize that he is observing merely a part of an operating totality, and that this totality, in turn, is but a small temporal segment of a personality” (Murray, 1938).

Closely related to the question of what the personality system’s elements are is the question of what, and how many, levels the system contains. Much like elements, there is no single answer. Rather, the number of levels depends on the research questions and the inferences one seeks to support.

For example, the study of population-level health aims to understand broad regularities and inform interventions that have aggregate effects across populations. In such cases, relatively coarse levels of analysis may be well aligned with the inferential goals. However, this level of

examination does not automatically permit inference about the person-level or within-person processes that generated those patterns. Explaining why such aggregate regularities arise would require explicitly specifying and modeling additional lower-order levels that connect population distributions to individual psychological dynamics.

Thus, the total number of levels that may exist in a personality system is not the same as the number of levels that must be represented in a given analysis. Pragmatically, not all theoretically possible levels need to be modeled. Yet reducing the number of levels is fundamentally different from ignoring levels that are necessary to support the inferences being made. Carefully bounding a system allows researchers to identify which levels can be provisionally treated as exogenous and which intermediary levels are indispensable for coherent explanation. The issue is therefore not whether a model contains many levels or few, but whether the specified hierarchy is sufficient to justify the inferential claims drawn from it (see Kuper et al., 2025), paralleling recent calls for explicit articulation of causal assumptions in psychological science (Rohrer, 2024; Wysocki et al., 2022).

From a pragmatic standpoint, beginning with a small number of levels, often no more than three, can provide a useful starting point. This is not because systems are inherently three-level structures, but because such a simplification aligns with existing modes of thinking (e.g., micro-, meso-, and macro-level phenomena), facilitates the use of common multilevel data structures (e.g., individuals nested within households and regions), and is increasingly supported by emerging analytic tools for estimating multi-level systems (Erofeeva et al., 2024). When additional levels are theoretically important, they can be layered onto these initial structures, allowing for progressively richer representations of complex systems.

5.3 How do we assess the system?

“The method required it, but the life did not” (Allport, 1968, p. 62).

Once the focal elements and levels of a system are specified, systems approaches carry clear implications for psychological measurement. First, assuming isomorphic structure across levels – that is, that validated between-person measures can be directly translated to within-person, momentary processes – implicitly assumes linear additivity across levels (Long, Lewis, et al., 2026). This assumption is incompatible with nonergodicity, self-organization, and emergence (Molenaar, 2004). Consequently, measures must be developed at the level of inference they are intended to capture. Simply adapting trait-level instruments for state-level assessment assumes linear additivity across levels and risks obscuring the heterogeneous dynamics that produce similar aggregate patterns.

Even when psychologically meaningful lower-order states can be specified, conventional self-report response formats may not provide neutral readouts of their magnitude. Recent psychophysical work indicates that bounded labeled scales are interpreted relative to person-specific internal benchmarks, such that identical endorsements may not reflect commensurate underlying experiences (Chituc & Scholl, 2025; Chituc et al., 2025). This benchmark dependence is consistent with a systems view in which higher-order person-level organization constrains the subjective evaluation of momentary states, suggesting that assessment must increasingly attend not only to target states, but to the evolving evaluative standards through which those states are interpreted and reported.

Second, the scope and timescale of measurement must be aligned with the level of the system (Batra, Johal, et al., 2025). Higher-order processes (e.g., cultural or contextual meaning systems) likely unfold over longer timescales and may be best captured through broader,

consensual assessments. In contrast, lower-level processes (e.g., momentary affective or cognitive dynamics) operate over shorter timescales and may require more fine-grained, person-specific measurement. However, these correspondences should be treated as heuristic rather than fixed as some higher-order processes may unfold quickly and some lower-level processes may operate slowly. Importantly, the appropriate timescale and breadth of measurement are not fixed, but are themselves empirical questions, requiring models that can adjudicate the levels at which psychological processes are organized.

If measures must be developed at the level of inference they are intended to support, then nonergodicity and heterogeneous person-specific organization imply that at least some aspects of assessment must shift from standardized top-down inventories toward partially personalized bottom-up approaches. Motivated by evidence for nonergodicity in personality structure and prediction, recent work has increasingly shifted toward personalized assessment (Büchner et al., 2025; Long, Büchner, et al., 2026; Modersitzki et al., 2025; Nissen & Beck, 2026). From a bottom-up perspective, this approach aims to rebuild measurement by starting with processes as they unfold within individuals and assessing the extent to which existing constructs capture these dynamics. In our own work, we have examined (1) the extent to which individuals' idiosyncratic self-descriptors are captured by existing trait and state measures (Figure 8A); (2) whether personalized assessments improve the prediction of momentary outcomes (Figure 8B); and (3) whether integrating these approaches can help map personality structure from the bottom up (Figure 8C).

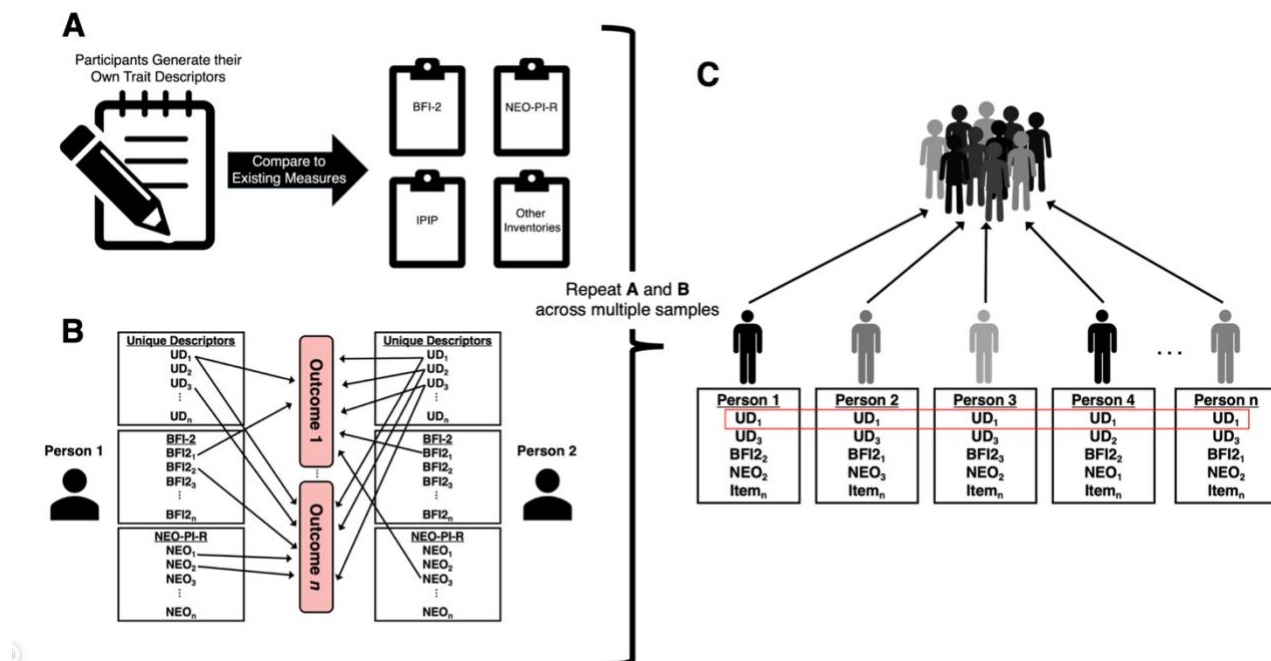


Figure 8. Using personalized assessment to capture between-person differences from the bottom up. (A) Taking a person’s self-descriptions and comparing content overlap with existing measures like the BFI-2 (Soto & John, 2017) and NEO-PI R (McCrae & Costa, 2008). See Long and colleagues (Long, Büchner, et al., 2026) for an example application of this approach. (B) Unique descriptors can then be compared to previous personalities inventories to see whether they are better predictors for momentary outcomes. Beck and Jackson (2022c) provide an example case with existing psychosocial measures while Büchner and colleagues (2025) and Nissen and Beck (2026) highlight the utility of unique descriptors for prediction. (C) Iterating across the procedures of A and B for multiple people can help identify commonalities in important indicators, dynamics, and personality structuring across people. This approach informs what is shared amongst people from the bottom-up by looking within the whole person, rather than standard assumptions that personality structures reflect presumed ergodic top-down. The section A Roadmap for Bottom-Up Integration in this chapter provides several considerations and future directions for incorporating bottom-up integration into practice.

Taken together, these considerations suggest that measurement is not a secondary concern, but central to how psychological systems are defined and understood. Crucially, within a systems framework, measurement is not merely the assessment of predefined constructs; it is the primary means by which the structure of the system is inferred. For example, in traditional latent variable models, highly correlated indicators are often treated as interchangeable reflections of a common underlying construct, motivating their aggregation to improve reliability (Borsboom et al., 2003; Uher, 2025). From a systems perspective, however, such correlations

may instead reflect components that are distinct but tightly coupled within the system. That is, the goal is not simply to measure variables, but to recover the organization of processes that give rise to them. From this perspective, advances in measurement are inseparable from advances in theory: Improving how we measure psychological phenomena is, in effect, improving our ability to identify the system itself (Chen et al., 2025; Flake & Fried, 2020).

These constraints collectively highlight both the promise and the challenge of a systems approach. In the following section, we outline a roadmap for addressing these challenges through bottom-up integration.

6 A Roadmap for Bottom-Up Integration

Personality psychology has long faced two seemingly antithetical tasks: Characterizing whole individuals in their uniqueness and capturing differences across people on shared dimensions (Cervone, 1991). We have argued that a systems approach requires placing the person, rather than the population, as the fundamental unit in psychological science. Under this view, however, the field's traditional solution to generalization – aggregating across individuals to derive common structure – becomes insufficient, as such aggregation risks obscuring the very person-specific organization one seeks to explain. Yet abandoning generalization altogether would preclude cumulative scientific progress. The remaining challenge, then, is how to reconcile idiographic, bottom-up insight with the development of generalizable knowledge.

Rather than treating these aims as incompatible, we suggest that the extent to which they can be integrated is itself an empirical question, one that must be approached bottom-up, from the person to the population (Figure 8C). At its core, this approach reframes a central question in psychological science: Not simply whether a theory or phenomenon is true, but for whom it is true, and under what conditions (Beck et al., 2025; Nissen & Beck, 2025b; Phan et al., 2025). By

treating person-specific models as units of analysis to be integrated rather than reduced, this framework makes it possible to address that question empirically, rather than assuming generality or treating heterogeneity as noise (Baars et al., 2025; Moeller, 2021).

Although our emphasis is on person-specific psychosocial dynamics, the logic of this approach is not confined to any single level of analysis. Comparable challenges arise whether one is studying biological processes, momentary psychological states, or broader behavioral patterns. In each case, the central issue is how to model organization at the level where processes unfold and then integrate upward without assuming unwarranted structural equivalence across levels (Anderson, 1972). Accordingly, while our illustrations focus on whole persons and populations, the inferential framework itself is intended to be level-agnostic.

We propose five steps for bottom-up integration, which are also depicted in Figure 9:

1. Define and model person-specific (lowest-level) dynamics.
2. Develop methods for comparing systems without reducing them to aggregate summaries.
3. Identify regularities that reflect shared organization or dynamics across systems while preserving person-specific structure.
4. Validate these regularities, including their stability over time and associations with meaningful outcomes.
5. Translate these insights into scalable approaches for population-level assessment.

To illustrate the first three steps, consider recent work on personalized models of well-being (Beck et al., 2025). Rather than adjudicating between top-down, bottom-up, or bidirectional theories of happiness at the group level, we estimated person-specific time-series networks linking domain-specific satisfaction (e.g., work, relationships) with life satisfaction (step 1). We then derived person-specific indices capturing whether individuals' well-being

dynamics were more top-down, bottom-up, or bidirectional (step 2), and examined their distribution and similarity across individuals (step 3). This allowed us to test whether individuals differ in the extent to which their well-being dynamics are better captured by different theoretical models.

This example reflects the first three steps of the proposed framework: Characterizing person-specific systems, developing metrics for comparing them, and identifying regularities across individuals without collapsing their structure. The remaining steps involve evaluating the stability and predictive validity of these patterns and developing scalable approaches for assessing such individual differences. We elaborate each step in turn.

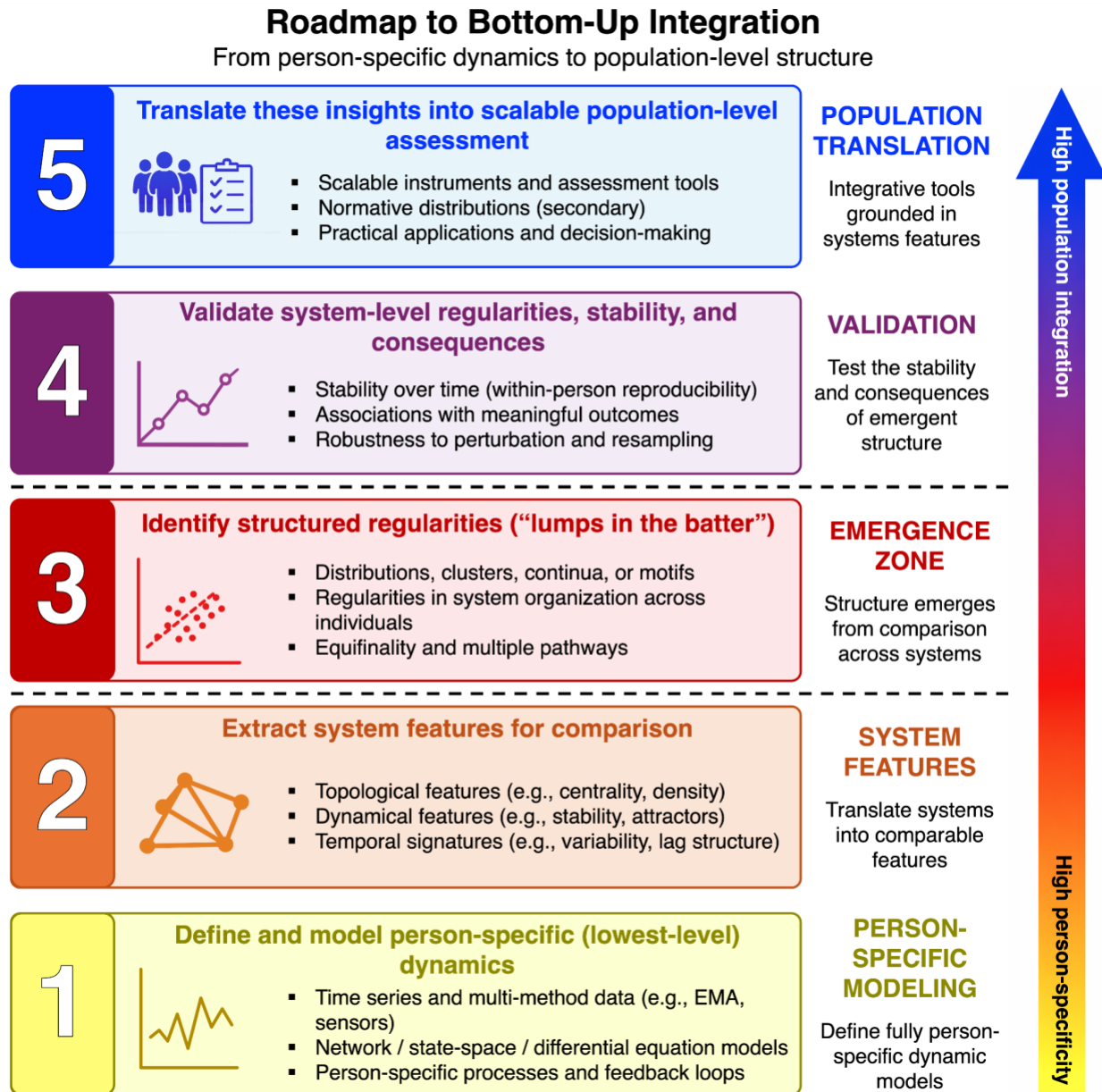


Figure 9. The figure depicts a five-stage framework for deriving population-level representations from idiographic systems. Beginning with person-specific dynamic modeling (Step 1), systems are translated into a comparable feature space that preserves their organization (Step 2). Cross-system comparisons yield structured heterogeneity ("lumps in the batter") reflecting recurring patterns without assuming homogeneity (Step 3). These emergent regularities are then evaluated for stability and external validity (Step 4), before being translated into scalable measurement approaches (Step 5). The vertical gradient reflects increasing integration and decreasing person-specificity, highlighting that population-level structure emerges from, rather than precedes, individual-level dynamics.

6.1 Step 1: Defining the Lowest Level System

The first step toward bottom-up integration requires building foundational models of the lowest level of interest – in our case, the phenomenologically defined, person-specific personality system. A central challenge at this stage is the temptation to model everything. As discussed in prior sections, however, capturing every component of the system is neither feasible nor necessary. Instead, this step involves identifying core components, guided by theory or inductive approaches, that are sufficient to represent the system at the level of interest.

This process is both difficult and foundational. More generally, it involves iteratively specifying system elements and their relationships, evaluating whether the resulting model captures expected patterns of behavior, and refining these representations in light of observed data (Chen et al., 2025). In some cases, this may involve formalizing system dynamics using tools such as differential equations or other computational models; in others, simpler statistical or machine learning approaches may suffice. The goal is not to recover a complete or final model of the system, but to arrive at a representation that is sufficiently specified to support inference at the level of interest. In this sense, model building is inherently iterative, with each refinement clarifying which components are essential for capturing the dynamics under study.

In ongoing work on personalized assessment, we initially approached this problem with minimal constraints. However, practical limitations, particularly assessment length, necessitated narrowing the scope. We therefore anchored the generation of personalized indicators to specific outcomes (e.g., procrastination, loneliness), alongside participant-selected outcomes, and paired these with candidate predictors drawn from prior research. This approach allows us to define person-specific systems under different conditions and examine processes such as multifinality and equifinality over time.

To evaluate whether these models capture distinct organizing forces, we compare alternative representations of the system, including multilevel and person-specific models, linear and nonlinear machine learning approaches, and dynamic systems models, as well as personalized versus standardized feature sets. These comparisons allow us to assess how different representations capture system dynamics and to identify which features are essential for modeling the system at the level of interest. Critically, these person-specific models serve as the foundation for subsequent steps, in which systems themselves become the units of comparison.

6.2 Step 2: Leveraging Tools for Capturing Dynamics and Organizing Principles

Once the system is defined, the next step is to summarize its dynamics without flattening or oversimplifying them. This step reflects a central feature of our approach: Person-specific models are not treated as endpoints but as inputs into a second layer of analysis in which systems themselves become the units of comparison. Rather than relying on aggregate summaries, this requires identifying features of systems that preserve their structure while enabling comparison across individuals. In practice, this often involves adapting existing tools in nonstandard ways to operate on person-specific models.

Across our work, we have taken several complementary approaches. One strategy involves extracting structural features of systems, such as network properties (e.g., centrality, connectivity patterns) that characterize how elements of the system are organized (Beck et al., 2025; Beck & Jackson, 2020a, 2021b; Nissen et al., under review; Nissen & Beck, 2025a). A second approach treats machine learning model performance as a summary of system predictability, using metrics such as accuracy, ROC AUC (area under the receiver operating curve), sensitivity, and specificity to capture how well different accounts for observed behavior predict unseen data (e.g., Beck & Jackson, 2022c). A third approach involves comparative

frameworks, including permutation tests, cross-validated model comparisons, and distance or profile similarity metrics, which allow systems to be compared without requiring them to be reduced to a common set of parameters (e.g., Beck et al., 2025; Beck & Jackson, 2022b; C. J. Lee & Beck, 2025). Finally, these approaches can be extended to the population level using meta-analytic summaries of system-derived features (S. A. W. Lee & Gates, 2024; Long & Beck, 2026a; Springstein et al., 2026), enabling inference while preserving heterogeneity.

This shift in the unit of analysis from observations to systems enables structured comparisons across individuals without collapsing their unique organization and is central to addressing questions of for whom particular system dynamics hold.

In our ongoing work on personalized assessment and prediction, we implement this logic using multiple complementary summaries. We evaluate model performance (accuracy, AUC, sensitivity, specificity) to assess predictability at the individual level, conduct cross-validated model comparisons to identify the best-performing feature sets for each person, and use variable importance metrics to characterize which features are most central to prediction. Together, these approaches capture different facets of system dynamics while preserving their underlying structure.

6.3 Step 3: Capturing Regularities in Dynamics and Organizing Principles

Summarizing person-specific systems is a necessary but insufficient step toward integration. The next challenge is to identify regularities in how these systems are organized, which Revelle (2021, personal communication) has described as “lumps in the batter.” These regularities reflect recurring patterns across individuals that may point to shared organizing principles of system dynamics.

Identifying such patterns requires moving beyond single summaries to integrative analyses across systems – that is, analyses that operate on features of person-specific models rather than aggregated observations. In practice, this can take several forms, all of which are heavily dependent on the output of Stage 2. Descriptive approaches, such as examining distributions, scatterplots, and correlations of system-level features, provide an initial sense of whether meaningful variability or clustering exists. More formal approaches, including clustering, multidimensional scaling, or latent profile analysis, can be used to identify subgroups of individuals with similar system dynamics, sometimes conceptualized as “ergodic subgroups,” in which system dynamics are more homogeneous within than across groups.⁴

At the same time, identifying regularities is not limited to unsupervised approaches. A complementary strategy involves linking system features to higher-level characteristics, such as sociodemographic or contextual factors, to test whether differences in system organization are systematically patterned. This approach treats higher-level variables not merely as covariates, but as potential organizing forces shaping system dynamics. From this perspective, Step 3 provides a first empirical foothold on the question of *for whom* systems differ: Identifying whether and how system organization is systematically patterned across individuals and their contexts.

Returning to our ongoing work, we begin with descriptive analyses of model performance across outcomes, feature sets, and modeling approaches to identify broad patterns. We then estimate multilevel models in which model performance (nested within individuals) is

⁴ Some approaches aim to do this a priori. For example, group iterative multiple model estimation (GIMME; Lane et al., 2016) is an algorithm used to estimate a series of person-specific models. Often estimated as networks using unified structural equation models, this approach first identifies pathways that improve person-specific models above some threshold (e.g., 75%). These pathways are then included (but freely estimated) in all person-specific models, along with additional pathways that improve model fit in the individual person-specific models. The subgrouping variant proceeds similarly, except that it first identifies subgroups, often using a walktrap algorithm and includes paths that improve most subgroup models.

predicted by model characteristics at Level 1 and sociodemographic and identity-based factors at Level 2. This allows us to test whether certain modeling approaches systematically perform better for particular groups. For example, we expect that individuals with minoritized identities may be less well captured by standard feature sets, consistent with the idea that dominant measurement frameworks may fail to capture the structure of their lived experiences (Nissen & Beck, 2026).

To further characterize these regularities, we examine similarity and divergence in system features directly. Distance-based approaches allow us to quantify absolute differences between individuals' systems, whereas profile-based approaches capture similarities in the patterning of features across systems (see e.g., C. J. Lee & Beck, 2025a). These complementary metrics provide a more nuanced account of both whether and how systems differ and can be similarly linked to sociodemographic predictors to test whether such differences are systematically patterned.

Critically, these approaches treat regularities not as assumptions, but as empirical questions, allowing shared structure to emerge from observed system organization rather than being imposed a priori. Together, these metrics allow us to distinguish whether systems differ in magnitude, organization, or both, and to test whether these differences are systematically associated with sociodemographic factors that may index underlying regulatory forces. Importantly, these analyses are not intended as tests of validity, but rather as tests of whether variability in system organization is structured and non-random across individuals.

6.4 Step 4: Validating Heterogeneity and Regularities Against External Criteria

Whereas Step 3 identifies for whom systems differ, Step 4 evaluates whether these differences are meaningful by testing whether variation in system organization predicts consequential outcomes, exhibits stability over time, and demonstrates discriminant validity.

In many ways, this step parallels traditional construct validation: The goal is to demonstrate that observed heterogeneity in system organization is reliable, stable, and consequential rather than arbitrary. However, because the focus is on features of person-specific systems rather than static scale scores, validation requires adapting standard approaches to operate at the level of system-derived features. Because system reliability cannot be established through traditional internal consistency approaches, such as coefficient alpha (Cronbach, 1951) or coefficient omega (McDonald, 1999), alternative techniques, including split-half or odd-even reliability, bootstrapping, and permutation tests, are more appropriate. Moreover, longitudinal and measurement-burst designs provide opportunities to evaluate stability over time, assessing whether features of person-specific systems exhibit temporal consistency (Beck & Jackson, 2020a).

Critically, validation in this context proceeds in at least two complementary ways. In some cases, validation is theory-driven, in which system features are evaluated against a priori expectations about their associations with outcomes. For example, in our work on links between emotion regulation goals and the dynamics of simple emotion systems, systems characterized by more goal-directed emotional dynamics support more effective regulation and, consequently, higher levels of well-being (Springstein et al., 2026). In such cases, validation involves testing whether person-specific system features align with these theoretically derived predictions.

In other cases, predictions emerge from the structure of the system itself. In our more exploratory work on dynamic networks (Beck & Jackson, 2022b; Nissen et al., under review; Nissen & Beck, 2025a), we draw on network-theoretic principles, including ideas from Cusp catastrophe theory, to generate predictions about system behavior. For example, more densely connected networks are expected to exhibit greater instability, such that small perturbations may produce disproportionately large changes (e.g., the vulnerability hypothesis; Cramer et al., 2016). From this perspective, features such as network density are not merely descriptive, but theoretically meaningful, predicting both short-term volatility in outcomes and the potential for longer-term personality change. Similarly, work using simple attractor-based representations evaluate whether system features capture stable and consequential dynamics (Büchner et al., 2026; C. J. Lee et al., 2026; Long & Beck, 2026b, 2026a; Springstein et al., 2026). As discussed in prior sections, these models also provide a formal framework for distinguishing dispositional tendencies from reactivity to perturbation; here, we leverage that framework as a theory-guided basis for validation, testing whether these features exhibit the expected patterns of stability and responsiveness.

Across both approaches, the goal is not simply to demonstrate that system features correlate with outcomes, but to establish that they have interpretable, theoretically grounded consequences. This includes evaluating predictive validity for future outcomes, stability across time, and discriminant validity from related constructs. More broadly, this step moves from identifying for whom systems differ to establishing whether those differences are consequential, providing evidence that variation in system organization is not only structured, but consequential.

6.5 Step 5: Translating System Insights Into Scalable Population-Level Assessment

The final step involves translating these insights into approaches that can be deployed at scale without sacrificing the fidelity of person-specific dynamics. This includes developing measurement strategies that efficiently capture key features of person-specific systems or their organizing principles, as well as tools that enable their use in prediction, classification, and intervention.

A central challenge at this stage is balancing the fidelity of person-specific dynamics with the practical constraints of large-scale research and application. Rich, intensive data, such as those collected through ecological momentary assessment or measurement-burst designs, provide unparalleled insight into system structure, but are (1) expensive, (2) time-intensive, and (3) often difficult to implement broadly. As such, a key goal of this step is to identify which features of person-specific systems are both informative and scalable, allowing for the development of novel measures that capture dynamic content and facilitate mapping between individual differences and underlying system dynamics.

One promising approach involves leveraging insights from person-specific models to inform the development of more efficient, partially standardized assessments. For example, in ongoing work on personality as a dynamic system, a core question is how many attractors characterize an individual's personality system. Leveraging multi-source data and cognitive interviews, Büchner and colleagues (2025) developed a novel procedure, the Personality Dynamics Inventory (PERDI), to capture this in cross-sectional surveys by asking participants to reconstruct their typical ways of being across behaviors and experiences selected by the researchers. A key implication is that this approach provides a scalable procedure for identifying

topological patterns without requiring long time series from a single individual. Such procedures can be readily implemented in ongoing research and large-scale longitudinal studies.

What distinguishes this approach from traditional measurement is that, rather than anchoring on broad scales and dimensions designed to differentiate individuals based on aggregate tendencies (often assuming these map onto underlying dynamics), it begins by establishing person-specific dynamics and then maps back to a cross-sectional procedure that best captures those dynamics. If these steps are executed successfully, the resulting measures of individual differences are no longer merely descriptive summaries, but theoretically grounded representations of person-specific system dynamics. In doing so, they enable person-level inferences that align with the underlying processes generating behavior, realizing a long-standing goal of psychological science that has remained difficult to achieve with aggregate, cross-sectional approaches.

More broadly, this step reframes integration not as the identification of universal effects, but as the development of tools that capture meaningful variation in system dynamics across individuals. From this perspective, scalable measurement does not require abandoning heterogeneity, but rather identifying principled, bottom-up ways to represent it efficiently. This includes developing hybrid approaches that combine personalized and standardized elements, as well as methods for classifying individuals based on shared system features, with the ultimate goal of creating measures of individual differences that map onto real heterogeneity in underlying dynamics.

Importantly, these efforts also provide a pathway toward personalized intervention. By identifying stable and predictive features of person-specific systems, and translating these features into scalable measurement tools, it becomes possible to tailor interventions to

individuals' system dynamics rather than relying on one-size-fits-all approaches. In this way, the same system-level features that support prediction and explanation can also serve as actionable targets for intervention.

6.6 Summary: Pathways for Future Bottom-up Research

Together, these steps outline a roadmap for building a cumulative science grounded in person-specific dynamics, in which understanding individuals is the primary step rather than a secondary refinement of population-level models. This perspective shifts the default assumption in psychological science from homogeneous effects to heterogeneous, person-specific processes. Although its application will vary across domains, the core principle remains the same: Psychological phenomena are best understood by first modeling the dynamics within individuals before identifying regularities across them. Rather than requiring new questions, this approach reframes existing ones, from top-down models that assume shared mechanisms to bottom-up approaches in which structure emerges from comparisons across individuals. Table 5 provides concrete entry points for this shift across common research traditions.

Table 5
Sample bottom-up research questions and approaches

Starting Point	Typical Top-Down Approach	Bottom-Up Reframing	First Step in Practice
Emotion (e.g., affect, mood)	Model average effects of or on emotion	Understand person-specific antecedents and construals of emotion	Estimate person-specific emotion dynamics before cross-person comparison
Clinical symptoms or diagnoses	Group individuals by symptom profiles	Treat symptoms as interacting components within person-specific clinical systems	Estimate person-specific symptom networks or temporal dynamics before identifying cross-person regularities
Well-Being	Identify average predictors of well-being	Understand unique determinants and experience of well-being	Model person-specific affect, context, and behavior dynamics before aggregation
Trait-like constructs	Assume stable latent traits explain behavior across individuals	Treat traits as emergent summaries of person-specific dynamics	Model person-specific dynamics and derive system-level features before population summaries

Situational reactivities	Model average situational effects across people	Model person-specific person-situation coupling	Estimate person-specific context-behavior relations across time
Structural factors	Treat structural factors as fixed average predictors	Treat structural factors as constraints on person-specific system dynamics	Model how structural conditions modulate person-specific processes and transactions
Development / change	Model average trajectories on common variables	Treat development as change in person-specific system topology	Track person-specific dynamics over time to detect system reorganization

7 Conclusion

Across the past century, personality psychology has made substantial progress in describing stable differences between individuals. Yet, as we have argued throughout this chapter, this progress has come with a fundamental limitation: The field has largely prioritized population-level regularities over the organization of psychological processes within persons. As a result, the dominant paradigm has become increasingly well-suited to describing how people differ, but less well-equipped to explain how personality operates.

In this chapter, we advanced an alternative perspective that personality is best understood as a person-specific, complex, nonlinear dynamic system. This claim is not intended as a rejection of prior work, but as a reorientation of explanatory priority. Traits, aggregate summaries, and population-level patterns remain valuable, but they are not foundational and often have limited explanatory utility. Rather, they are outcomes of underlying systems that unfold within individuals over time. From this perspective, the central task of personality science is not to hasten to generalization, but to hasten to the internal pattern.

Adopting this perspective has several implications. Conceptually, it shifts the unit of analysis from variables to the person system and reframes personality as an emergent, multi-level process shaped by interacting regulatory forces. Formally, it highlights the limits of population-level inference under conditions of nonergodicity and underscores the need for approaches that align with the dynamic, heterogeneous nature of psychological phenomena. Empirically, it calls

for methods that model person-specific dynamics, compare systems without reducing them to common summaries, and identify regularities through bottom-up integration. As we outlined in our roadmap, this involves moving from defining person-specific systems, to comparing them, identifying shared organizing principles, validating their consequences, and ultimately translating these insights into scalable approaches for assessment and intervention.

Importantly, this framework does not eliminate the goal of building cumulative, generalizable knowledge. Instead, it reframes that goal as one of integration rather than aggregation. *General principles do not precede the study of individuals; they emerge from it.* This shift carries both scientific and practical implications. Scientifically, it provides a pathway for resolving longstanding tensions between idiographic and nomothetic approaches. Practically, it opens the door to more precise prediction and more effective, personalized interventions by grounding inference in the dynamics that generate behavior within individuals.

We recognize that many of the approaches described here remain in early stages of development. The tools, data, and theoretical frameworks required to fully realize this vision are still emerging. However, the central claim is not contingent on the maturity of any particular method. Rather, it follows from a basic alignment between the phenomena we seek to explain and the frameworks we use to explain them. If personality resides within persons as a dynamic, organized system, then our theories, methods, and inferences must be built accordingly.

Ultimately, the question that has long haunted personality psychology – what makes the system cohere in any one person – cannot be answered by looking only at populations. It requires returning to the individual, not as a noisy instance of a general pattern, but as the locus of the phenomena themselves. Only from that starting point can the field begin to discover how broader psychological structure emerges.

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